Horizon 2020

Call: H2020-MSCA-IF-2016
(Marie Skłodowska-Curie Individual Fellowships)

Topic: MSCA-IF-2016

Type of action: MSCA-IF-EF-SE
(Society and Enterprise panel)

Proposal number: 752907

Proposal acronym: DATAMINE4.0

Deadline Id: H2020-MSCA-IF-2016

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How to fill in the forms?

The administrative forms must be filled in for each proposal using the templates available in the submission system. Some data fields in the administrative forms are pre-filled based on the previous steps in the submission wizard.

This proposal version was submitted by **Arnau FATJO** on **14/09/2016 14:55:23** Brussels Local Time. Issued by the Participant Portal Submission Service.

1 - General information

Topic	MSCA-IF-2016
Call Identifier	H2020-MSCA-IF-2016
Type of Action	MSCA-IF-EF-SE
Deadline Id	H2020-MSCA-IF-2016

Acronym	DATAMINE4.0
Proposal title	Advanced Data Modeling and Analysis Applied to next Generation Industry 4.0 settings and the Internet of Things

Note that for technical reasons, the following characters are not accepted in the Proposal Title and will be removed: < > " &

Duration in months	24
Scientific Area	ENG

Please select up to 5 descriptors (and at least 3) that best characterise the subject of your proposal, in descending order of relevance.

Descriptor 1	*Numerical analysis, simulation, optimisation, modelling tools, data mining*	
Descriptor 2	*Scientific computing and data processing*	
Descriptor 3	*Artificial intelligence, intelligent systems, multi agent systems*	Remove
		Add

Free keywords	*Data Mining, Industrial Process Optimisation, Data Analysis, Network Analysis, Data Fusion, Graph Analysis*

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Proposal ID **752907** *Acronym* **DATAMINE4.0**

Abstract

DATAMINE4.0 is an IF for Career Development for Natasa Sarafijanovic-Djukic, PhD in Computer/Communication Sciences obtained from EPFL, Switzerland, a researcher in statistical modelling originally from Serbia, whereby she will work in IRIS, a Spanish R&D and advanced engineering SME. The project will study the modeling and analysis of data coming from complex industrial and organizational applications, in order to identify hidden causistic relationships in order to calibrate production and organizational control systems. It is highly multidisciplinary combining data science, statistics, artificial intelligence, telecommunications, specific engineering and organizational domains and information technology processing competences. In line with new paradigms such as Industry 4.0 and the Internet of Things. Although machine learning paradigms have been in existence for several decades, their penetration in the industrial and public organization setting has been limited. The goal is to build calibration and optimization models using advanced statistical and machine learning methods and other artificial intelligence inspired algorithms, for applications in which existing methods have been unsatisfactory. This will have a big impact on productivity and repeatability of winning formulas for key industrial applications such as customized production, public sector applications (e.g. optimization of urban mobility), and food processing. This is in line with findings published by the European Union. The developed data mining solutions will greatly enhance the current performance of the selected applications, allowing them a competitive advantage over current methods. The work will be done in an applied R&D environment in line with the objective for Dr. Natasa Sarafijanovic-Djukic's career continuation. An engineering approach will be ideally completed with 7 formal scientific trainings, including two academic secondments, each of 2 months.

Remaining characters 38

Has this proposal (or a very similar one) been submitted to a Horizon 2020 Marie Skłodowska-Curie Individual Fellowship call? ○ Yes ◉ No

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Declarations

1) The applicant (future beneficiary) declares to have the explicit consent of all partner organisations (if applicable) on their participation and on the content of this proposal.	☒
2) The information contained in this proposal is correct and complete.	☒
3) This proposal complies with ethical principles (including the highest standards of research integrity — as set out, for instance, in the European Code of Conduct for Research Integrity — and including, in particular, avoiding fabrication, falsification, plagiarism or other research misconduct).	☒

4) The applicant (future beneficiary) confirms:

- to have carried out the self-check of the financial capacity of the organisation on https://ec.europa.eu/research/participants/portal/desktop/en/organisations/lfv.html or to be covered by a financial viability check in an EU project for the last closed financial year. Where the result was "weak" or "insufficient", the applicant (future beneficiary) confirms being aware of the measures that may be imposed in accordance with the H2020 Grants Manual (Chapter on Financial capacity check); or	◉
- is exempt from the financial capacity check being a public body including international organisations, higher or secondary education establishment or a legal entity, whose viability is guaranteed by a Member State or associated country, as defined in the H2020 Grants Manual (Chapter on Financial capacity check); or	○
- as sole participant in the proposal is exempt from the financial capacity check.	○

5) The applicant (future beneficiary) hereby declares:

- it is fully eligible in accordance with the criteria set out in the specific call for proposals; and	☒
- it has the financial and operational capacity to carry out the proposed action.	☒

The applicant (future beneficiary) is only responsible for the correctness of the information relating to his/her own organisation. Where the proposal to be retained for EU funding, the applicant (future beneficiary) will be required to present a formal declaration in this respect.

According to Article 131 of the Financial Regulation of 25 October 2012 on the financial rules applicable to the general budget of the Union (Official Journal L 298 of 26.10.2012, p. 1) and Article 145 of its Rules of Application (Official Journal L 362, 31.12.2012, p.1) applicants found guilty of misrepresentation may be subject to administrative and financial penalties under certain conditions.

Personal data protection

Your reply to the grant application will involve the recording and processing of personal data (such as your name, address and CV), which will be processed pursuant to Regulation (EC) No 45/2001 on the protection of individuals with regard to the processing of personal data by the Community institutions and bodies and on the free movement of such data. Unless indicated otherwise, your replies to the questions in this form and any personal data requested are required to assess your grant application in accordance with the specifications of the call for proposals and will be processed solely for that purpose. Details concerning the processing of your personal data are available on the privacy statement. Applicants may lodge a complaint about the processing of their personal data with the European Data Protection Supervisor at any time.

Your personal data may be registered in the Early Warning System (EWS) only or both in the EWS and Central Exclusion Database (CED) by the Accounting Officer of the Commission, should you be in one of the situations mentioned in:
-the Commission Decision 2008/969 of 16.12.2008 on the Early Warning System (for more information see the Privacy Statement), or
-the Commission Regulation 2008/1302 of 17.12.2008 on the Central Exclusion Database
(for more information see the Privacy Statement) .

Proposal ID **752907** *Acronym* **DATAMINE4.0**

List of participants

#	Participant Legal Name	Country
1	INNOVACIO I RECERCA INDUSTRIAL I SOSTENIBLE SL	Spain

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2 - Administrative data of participating organisations

Future Host Institution

PIC	***Legal name***
998993786	*INNOVACIO I RECERCA INDUSTRIAL I SOSTENIBLE SL*

Short name: IRIS

Address of the organisation

Street AVENIDA DEL CANAL OLIMPIC 15 PARC MED

Town CASTELLDEFELS

Postcode 08860

Country Spain

Webpage www.iris.cat

Legal Status of your organisation

Research and Innovation legal statuses

Public body ..... no

Non-profit ..... no

International organisation ..... no

International organisation of European interest ..... no

Secondary or Higher education establishment ..... no

Research organisation ..... no

Small and Medium-sized Enterprises (SMEs) ..... yes

Academic Sector ..... no

Legal person ..... yes

NACE Code: 721 - Research and experimental development on natural sciences and engineering

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Department(s) carrying out the proposed work

Department 1

Department name: Data Science

☐ not applicable

☒ Same as organisation address

Street: AVENIDA DEL CANAL OLIMPIC 15 PARC MEDITE

Town: CASTELLDEFELS

Postcode: 08860

Country: Spain

If the location of the Department carrying out the proposed work is not the same as the location of the Host Institute, please note that although the proposal submission system calculates the budget of the project based on the location of the Host Institute, the budget of the project for the grant agreement will be calculated by using the country coefficient of the location of the Department carrying out the proposed work.

Researcher

The name and e-mail of the Researcher and Supervisor are read-only in the administrative form, only additional details can be edited here. To give access rights and contact details of contact persons, please go back to Step 4 of the submission wizard and save the changes.

Researcher ID	*If you have a researcher identifier number (e.g. ResearcherID, ORCID) please enter it here.*		
Last Name*	Sarafijanovic-Djukic	Last Name at Birth	
First Name(s)*	Natasa	Gender*	○ Male ◉ Female
Title	Dr.	Country of residence*	Switzerland
Nationality*	Serbia	Nationality 2	
Date of Birth (DD/MM/YYYY)	15/09/1974	Country of Birth*	Serbia
		Place of Birth	Cacak

Contact address

Current organisation name	-		
Current Department/Faculty/Institute/ Laboratory name	-		
	☐ Same as organisation address		
Street	Tobelhof 17		
Postcode/Cedex	8134	Town	Adliswil
Phone	+41445594232	Country	Switzerland
Phone2 / Mobile	+41764464381		
E-Mail*	natasa.sarafijanovic@gmail.com		

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Qualifications

University Degree	Date of award (DD/MM/YYYY)	31/01/2000
Doctorate (in progress)	Date of award (DD/MM/YYYY)	
Doctorate	Date of award (DD/MM/YYYY)	23/02/2010
Full time postgraduate research experience	Number of months	
Other Academic qualifications	Date of award (DD/MM/YYYY)	

Place of activity/place of residence (previous 5 years - most recent one first)

Indicate the period(s) and the country/countries in which you have legally resided and/or had your main activity (work, studies, etc) during the last 5 years up until the deadline for the submission of the proposal. Please fill in this section without gaps, until the call deadline (14/09/2016).

Period from	Period to	Duration (days)	Country
02/10/2002	14/09/2016	5097	Switzerland
	Total	5097	

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Proposal ID **752907** *Acronym* **DATAMINE4.0** *Short name* **IRIS**

Supervisor

The name and e-mail of the Researcher and Supervisor are read-only in the administrative form, only additional details can be edited here. To give access rights and contact details of contact persons, please go back to Step 4 of the submission wizard and save the changes.

Title: Dr.

Sex: (•) Male ( ) Female

First name*: **David**

Last name*: **Nettleton**

E-Mail*: **david.nettleton@iris.cat**

Position in org.: Data Scientist

Department: Data Science

☒ Same as organisation address

Street: AVENIDA DEL CANAL OLIMPIC 15 PARC MEDITERRANI DE LA TECNOLO

Town: CASTELLDEFELS

Post code: 08860

Country: Spain

Website: http://www.iris-eng.com/

Phone: +34935542505

Phone 2: +xxx xxxxxxxxx

Fax: +xxx xxxxxxxxx

Other contact persons

First Name	Last Name	E-mail	Phone
Arnau	FATJO	arnau.fatjo@iris.cat	+34935542512
Elodie	Bugnicourt	ebugnicourt@iris.cat	
Albert	Torres	albert.torres@iris.cat	

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3 - Budget

Is the Researcher eligible for family allowance? (•) Yes ( ) No

Participant Number	Organisation Short Name	Country	Country Coefficient	Number of Months	Researcher Unit Cost			Institutional Unit Cost		Total
					Living Allowance	Mobility Allowance	Family Allowance	Research, training and networking costs	Management and Overheads	
1	IRIS	ES	0,976	24	108921,60	14400,00	12000,00	19200,00	15600,00	170121,60
Total				24	108921,60	14400,00	12000,00	19200,00	15600,00	170121,60

Partner Organisation from Third Country does not sign the Grant Agreement, does not recruit the researcher and does not directly claim costs from the action. The entire EC contribution is transmitted to the Host organisation located in Members States or Associated Countries.

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4 - Ethics issues table

1. HUMAN EMBRYOS/FOETUSES		Page
Does your research involve Human Embryonic Stem Cells (hESCs)?	○ Yes ◉ No	
Does your research involve the use of human embryos?	○ Yes ◉ No	
Does your research involve the use of human foetal tissues / cells?	○ Yes ◉ No	
2. HUMANS		**Page**
Does your research involve human participants?	○ Yes ◉ No	
Does your research involve physical interventions on the study participants?	○ Yes ◉ No	
3. HUMAN CELLS / TISSUES		**Page**
Does your research involve human cells or tissues (other than from Human Embryos/ Foetuses, i.e. section 1)?	○ Yes ◉ No	
4. PERSONAL DATA		**Page**
Does your research involve personal data collection and/or processing?	○ Yes ◉ No	
Does your research involve further processing of previously collected personal data (secondary use)?	○ Yes ◉ No	
5. ANIMALS		**Page**
Does your research involve animals?	○ Yes ◉ No	
6. THIRD COUNTRIES		**Page**
In case non-EU countries are involved, do the research related activities undertaken in these countries raise potential ethics issues?	○ Yes ◉ No	
Do you plan to use local resources (e.g. animal and/or human tissue samples, genetic material, live animals, human remains, materials of historical value, endangered fauna or flora samples, etc.)?	○ Yes ◉ No	
Do you plan to import any material - including personal data - from non-EU countries into the EU?	○ Yes ◉ No	
Do you plan to export any material - including personal data - from the EU to non-EU countries?	○ Yes ◉ No	
In case your research involves low and/or lower middle income countries, are any benefits-sharing actions planned?	○ Yes ◉ No	
Could the situation in the country put the individuals taking part in the research at risk?	○ Yes ◉ No	

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7. ENVIRONMENT & HEALTH and SAFETY		Page
Does your research involve the use of elements that may cause harm to the environment, to animals or plants?	◯ Yes ◉ No	
Does your research deal with endangered fauna and/or flora and/or protected areas?	◯ Yes ◉ No	
Does your research involve the use of elements that may cause harm to humans, including research staff?	◯ Yes ◉ No	
8. DUAL USE		**Page**
Does your research involve dual-use items in the sense of Regulation 428/2009, or other items for which an authorisation is required?	◯ Yes ◉ No	
9. EXCLUSIVE FOCUS ON CIVIL APPLICATIONS		**Page**
Could your research raise concerns regarding the exclusive focus on civil applications?	◯ Yes ◉ No	
10. MISUSE		**Page**
Does your research have the potential for misuse of research results?	◯ Yes ◉ No	
11. OTHER ETHICS ISSUES		**Page**
Are there any other ethics issues that should be taken into consideration? Please specify	◯ Yes ◉ No	

I confirm that I have taken into account all ethics issues described above and that, if any ethics issues apply, I will complete the ethics self-assessment and attach the required documents. ☒

How to Complete your Ethics Self-Assessment

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5 - Call specific questions

Eligibility Researcher (future fellow)

1. Were you in the last 5 years in military service? ○ Yes ◉ No

Other Questions

For communication purposes only, the REA asks for permission to publish the name of the researcher (future fellow) should the proposal be retained for funding.

1. Does the researcher (future fellow) give this permission? ◉ Yes ○ No

2. Is there a secondment in Member States or Associated Countries envisaged in Part B of this proposal? ◉ Yes ○ No

In which sector is the secondment in Member States / Associated Countries foreseen?

☒ Academic ☐ Non Academic

Do you already know the organisation to which this secondment will be? ◉ Yes ○ No

Name: KU Leuven

Country: Belgium

In which sector is the secondment in Member States / Associated Countries foreseen?

☒ Academic ☐ Non Academic

Do you already know the organisation to which this secondment will be? ◉ Yes ○ No

Name: University of Skövde

Country: Sweden

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Data management activities

A new focus within Horizon 2020 is data management, for example through the use of Data Management Plan (DMP).

DMPs detail what data the project will generate, whether and how it will be exploited or made accessible for verification and re-use, and how it will be curated and preserved.

The use of a DMP is required for projects participating in the Open Research Data Pilot in the form of a deliverable in the first 6 months of the project (possible updates during the project).

Other projects are invited to submit a DMP if relevant for their planned research.

Are data management activities relevant for your proposed project?	○ Yes	◉ No

Open Research Data Pilot in Horizon 2020

All applicants can participate in the Pilot on Open Research Data in Horizon 2020[1] on a voluntary basis. This Pilot aims to improve and maximise access to and re-use of research data generated by actions.

Participants in the Pilot will be invited to formulate a Data Management Plan (DMP). DMPs detail what data the project will generate, whether and how it will be exploited or made accessible for verification and re-use, and how it will be curated and preserved.
Participating in the Pilot is flexible in the sense that it does not mean that all research data needs to be open. Rather, projects can define certain datasets to remain closed via a Data Management Plan (DMP).

Please note that participation in this Pilot does not constitute part of the evaluation process. Proposals will not be evaluated favourably because they participate in the Pilot on a voluntary basis.

We wish to participate in the Pilot on Open Research Data in Horizon 2020 on a voluntary basis	○ Yes	◉ No

[1] *According to article 43.2 of Regulation (EU) No 1290/2013 of the European Parliament and of the Council, of 11 December 2013, laying down the rules for participation and dissemination in "Horizon 2020 - the Framework Programme for Research and Innovation (2014-2020)" and repealing Regulation (EC) No 1906/2006.*

STARTPAGE

MARIE SKLODOWSKA-CURIE ACTIONS

Individual Fellowships (IF)
Call: H2020-MSCA-IF-2016

PART B

"DATAMINE4.0"

This proposal is to be evaluated as:
Standard EF

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LIST OF PARTICIPANTS

Participants	Legal Entity Short Name	Academic (tick)	(tick)Non-academic	Country	Dept,/ Division/ Laboratory	Supervisor	Role of Partner Organisation
Beneficiary	IRIS		x	Spain	Engineering Group	Dr. David F. Nettleton	Host
Partner Organisation	University KU Leuven	x		Belgium	Department of Computer Science	Prof. Jesse Davis, Dr. Anton Dries	training of the fellow regarding graph matching
Partner Organisation	University of Skövde	x		Sweden	Department of Computer Science	Prof. Vicenc Torra	training of the fellow regarding data fusion

Data for non-academic beneficiaries

Name	Location of research premises (city/country)	Type of R&D activities	No. of employees	No. of employees in R&D	Web site	Annual turnover (approx. in Euro)	Enterprise status (Yes/No)	SME status (Yes/No)
Innovacio i Recerca industrial i sostenible (IRIS)	Mediterranean Technology Park, Avda. Carl Friedrich Gauss nº 11 08860 Castelldefels (Barcelona) - Spain	Industrial transfer, environmental optimization, development and processing of materials, food, etc.	48		http://iris.cat/		Yes	Yes

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1. EXCELLENCE

1.1. Quality and credibility of the research/innovation action (level of novelty, appropriate consideration of inter/multidisciplinary and gender aspects)

1.1.1. Introduction, state-of-the art, objectives and overview of the action

Although data mining has been in existence for several decades, its application to industrial settings and public organizations has been limited. In recent years new more robust methods such as kernel based algorithms have become sufficiently reliable to be used for critical in-line settings such as high-frequency financial trading, PLC monitoring of industrial processes, logistics of public urban movement (trains, bus, metro) and the incorporation of mobile device technologies (eg. RFID), among others. However, the success in deploying such solutions depends, apart from the algorithms themselves, on a skilful analysis of the environment to be modeled, knowledge and data representation, data pre and post-processing (including input factor selection based on data relevance and reliability), choice of techniques, and so on. The choice of data mining paradigms can be problem and data dependent. We can contrast different learning paradigms, from the "best of breed", for example, among the **top ten algorithms in data mining** as indicated in the highly referenced paper by Zhou et al.[3]: C4.5[5], k-Means, SVM[6], Apriori, EM, PageRank, AdaBoost, kNN, Naive Bayes[4], and CART. This list has remained remarkably static since it was published in 2007, however more recent lists have added new more specific variants:

- kdnuggets poll (2011): methods: Decision Trees/Rules 59.8 %, Regression 57.9 %, Clustering 52.4 %, Statistics (descriptive) 47.9%, http://www.kdnuggets.com/polls/2011/algorithms-analytics-data-mining.html
- kdnuggets poll (2015): Linear regression; Logistic regression; k-means; SVMs; Random Forests; Matrix Factorization/SVD; Gradient Boosted Decision Trees/Machines, ...

In the specific context of industrial process modeling, neural networks and regression are popular methods, with their respective specific variants. For example, a Linear Quadratic Gaussian (LQG) subspace technique was used[7] to train a supervised learner to control the tension of a winding process. A second empirical study[8], modeled the following industrial pullwinding process to control the friction force of a micro-pullwinding process. The objectives were (i) to train a model to predict the friction and (ii) to perform a calibration step to find the input values which result in a given friction. The Industry 4.0 framework[1,2], in which the incorporation of "intelligent" techniques in industrial processes is a central theme, is expected to give a strategic impulse to the use of advanced data mining techniques to model the data generated by and which describes the in-line production processes. In the light of the above, the overall aim of this DATAMINE4.0 project is to study real complex industrial processes, execute data modeling and calibration projects and deploy them in situ, wherever possible. The following scientific objectives are proposed to significantly advance the state of the art to deliver scientific results that are useful in real industrial settings:

- Review the state of the art for statistical, machine learning/calibration methods as solutions for the Industry4.0 framework (more specific details will also be studied once case studies are selected).
- To select and execute data mining projects, with special emphasis on Industry4.0 solutions, integrating the needs of end users and the entire production chain in order to define market requirements. To perform "data audit" studies in order to obtain all the information necessary for the data mining projects.
- To execute data mining projects "data project".
- To perform the characterization of new data modeling vs. current, generally empirical, process optimisation data management approaches used in industrial sectors.
- To develop profitable optimized control and parametrization of industrial processes, thus promoting the Industry 4.0 paradigm as well as incorporating machine learning and AI data modeling techniques into mobility and other applications.
- To study applications within the "Internet of Things" (IOT) and Cloud Computing paradigms, processing data from such technologies as RFID and other mobile devices, wireless peer-to-peer communications for logistics and real time tracking as dynamic network analysis. This also includes sensor data fusion.
- To study the critical processing aspects to plan the transfer of the results to industrial lines and further in-situ validation to demonstrate the potential of DATAMINE4.0 to significantly improve the operations of the respective end users.

The developed industrial process data models present the basis for a high performance new process modeling with excellent calibration properties that will gain wide adoption over time. Scientific knowledge and technology transfer competitiveness underpins all the themes, including new manufacturing and resulting business models that are essential for the realization of value. The project will be a significant step forward for data modeling and calibration on in-situ industrial processes. In Fig. 1, we see a typical development cycle to build a data model and calibrate[9,10,11,12] the inputs for a given output. Different data modeling techniques can be used for optimization problems of this type, such as non-linear regression[10,11], neural networks[14], Bayesian Models[15] and genetic algorithms[13], among others.

1.1.2 Research methodology and approach

A portfolio of successful data mining models will be developed using a two step process[16] in which Step 1 involves conducting several "data audits" each of which will define the requirements of a data mining project in terms of its business objective(s) and by Step 1. The "data audit" step requires first the definition of a viable business objective which may be to solve a current production problem or optimization challenge of an industrial process, for example.

1. Davies, R., Industry 4.0 - Digitalisation for productivity and growth, European Parliament Briefing Sept. 2015, European Parliamentary Research Service, PE 568.337 2. Butter, M., Fischer, N., Gijsbers, G., Hartmann, C., de Heide, M., van der Zee, F., Horizon 2020: Key Enabling Technologies (KETs), Booster for European Leadership in the Manufacturing Sector (2014), Study for the ITRE Committee, European Parliament Directorate General for Internal Policies, IP/A/ITRE/2013-01 October 2014, PE 536.282 3. Yu, S., Zhou, Z.H., Steinbac, M., Hand, D.J., Steinberg, D. (2007) Top 10 algorithms in data mining. Knowl Inform Syst 14(1):1–37. 4. John, G.H., Langley, P. (1995) Estimating continuous distributions in Bayesian classifiers. In: 11th Conf. on Uncertainty in Artificial Int., pp 338–345. 5. Quinlan, J.R. (1993) C4.5 programs for machine learning. Morgan Kaufmann. 6. Vapnik, V.N. (1995) The nature of statistical learning theory. Springer Verlag. 7. Bastogne, T., Noura, H., Richard, A., Hittinger, J.M., Application of subspace methods to the identification of a winding process. In: Proc. of the 4th European Control Conference, Vol. 5, 1-4 July 1997, Brussels. 8. Nettleton, D.F., Bugnicourt, E., Wasiak, C., Rosales, A. Towards Automatic Calibration of In-Line Machine Processes, 18th International Conference on Industrial Engineering and Manufacturing Process Modeling, ICIEMPM 2016, London, United Kingdom, July 28-29, 2016.

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This will involve a "consultancy study" of the necessary data in order to achieve the goal. Then Step 2 "data project" is the execution and deployment of the data mining project defined through a series of meetings with the stakeholders (Management, IT Department, Domain Experts) in order to consensuate the business objective and identify alternatives if necessary. For example, a viable business objective would be one in which a significant improvement is possible with respect to the current performance. Once this is done, an information and data capture exercise is performed in order to evaluate if there exists sufficient quality data in order to analyze and model the business objective which has been identified.

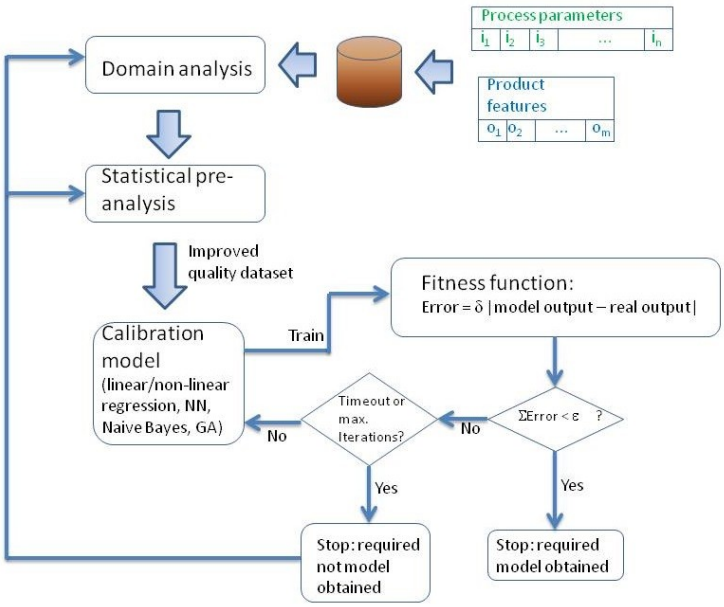

Fig. 1 Example calibration of a data model

The "data project" step follows the typical phases of a data mining project, assuming that at least part of the work of data identification and variable selection has been done in the "data audit" step. (i) Test data sets are constructed (which can be fused from diverse data sources such as Excel files, relational databases, text corpus, flat files, etc.) and these are analyzed in order to derive insights which enable the definition of new higher information value factors. This can involve standard statistical techniques such as correlation, principal components, information gain, k-Means clustering, tree-maps, hierachical descomposition clustering, rule induction, association analysis, among others. (ii) data models are built based on the data analysis performed in step (i) using contrasting algorithmic paradigms, such as regression models (linear, non-linear, logistic), "black box" supervised models (neural networks (e.g. feedforward, recurrent), support vector machines with different kernels (e.g. radial basis function); "white box" techniques[17,18] such as rule induction (e.g. MPART, J48, ID3, RandomForest), Bayesian classifers; other supervised techniques, such as Ibk. With respect to Model Calibration, we propose to make some improvements to the Monte Carlo type method in which Gaussian random numbers are generated for each input: firstly by "remembering" the best solution found to date during the generation of the different combinations; secondly by benchmarking Particle Swarm Optimization which has already indicated promising results for an industrial calibration application, and represents a family of artificial intelligence paradigms inspired in natural processes, which uses heuristics and rules to constantly guide the processing towards an optimal solution.

1.1.3. Originality and innovative aspects of the research programme

Aiming **to open up new fields in the modeling of industrial processes with a holistic approach**, this project involves unique challenges and a high level of innovation to achieve **major and pressing objectives for industrial manufacturing companies.** *Such companies* must balance customization with volume in-line production with such issues as machine calibration, translating customer requirements definition into machine production instructions, time series prediction, among others. The lack of a systematic approach and the required barrier improvements justify further research to develop suitable methodologies and build a portfolio of successful industrial data modeling deployments in different sectors. Also lacking in the prior art, development of precise "white box" rule based data models which complement the traditional "black box" statistical and supervised learning algorithms, and generic parametric solutions which can be customized for different scenarios. This last aspect is contrasted with the current state of art in which specific non-portable modeling solutions are typically developed to solve specific problems.

DATAMINE4.0 expects **to make a major breakthrough to allow the transfer of results to industrial production lines or urban environment, for end users** of initially (a) food processing, (b) urban mobility and (c) customized production of composite materials components, but potentially also in other scenarios including telecommunications technologies to take benefit from the fellow's related expertise. Within the **Internet of Things (IoT)** paradigm, this will englobe data coming from mobile devices, wireless peer-to-peer communications and RFID, among others. Those initial case scenario have been selected in view of: (i) the number of target end users in such sectors, where IRIS generally already has expertise in terms of providing other engineering or monitoring services, leading to synergies and indeed a high potential for IRIS when tackling the post-project business replication of the project results; (ii) their representability of a greater number of case scenarios which may be transferred to other sectors; (iii) the need of different data mining approaches to cover them leading to a rich training experience for the fellow; (iv) their having a high optimisation potential when taking full benefits of the data they generate. Thus, following the work plan the proposal will:

- **Identify and define viable data mining projects** to focus time and effort on business objectives and/or proposals with sufficient data availability and/or data quality in the **industrial production and urban mobility fields.**

9. Martens, H., Naes, T., Multivariate Calibration. Wiley 1989, pp. 79-82. 10. Fitting Linear Models JMP 12 (SAS), 2015. 11. Lavagnini, I., Magno, F., A STATISTICAL OVERVIEW ON UNIVARIATE CALIBRATION, INVERSE REGRESSION, AND DETECTION LIMITS: APPLICATION TO GAS CHROMATOGRAPHY/MASS SPECTROMETRY TECHNIQUE, Mass Spectrometry Reviews, 2007, 26,1–18. 12. Lyman Ott, R., Longnecker, M., An Introduction to statistical methods and data analysis. Brokks/Cole, 2010 (6th edition), pp. 604-607, "the inverse regression problem" (calibration). 13. Yazdani, M., Daryabari, A., Farshi, A., Talatahari, S., Application of Taguchi Method and Genetic Algorithm for Calibration of Soil Constitutive Models, Journal of Appl. Math., Vol. 2013 (2013), Artic. ID 258721. 14. Despagne, F., Luc Massart, D., Neural networks in multivariate calibration. Analyst, 1998, 123, 157R–178R 15. Prudhomme, S., Bayesian Inversion for Model Calibration, RTO-AVT-VKI short course on Uncertainty Quantification Stanford University, California, April 15-16, 2011. 16. Nettleton, D.F., Commercial Data Mining - Processing, Analysis and Modeling for Predictive Analytics Projects, 1st Edition (2014), Morgan Kaufmann Publishers, ISBN 9780124166028, 304 pages. 17. SIMCA (Soft Independent Modeling of Class Analogies), https://en.wikipedia.org/wiki/Soft_independent_modelling_of_class_analogies. 18. Shmueli, G., To Explain or to Predict? Statistical Science 2010, Vol. 25, No. 3, 289–310.

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- **Execute the viable projects defined in the previous step,** involving data selection, refinement, modeling and calibration, choosing the most adequate analysis tools for each task.
- Address the calibration of industrial production composed of complex interrelated multiple sub-processes to **create parametrizable customized mass production**
- With reference to calibration, experiment with new paradigms such as guided Gaussian generators and Particle Swarm Optimization.
- **Deploy the models** built in the previous step in real industrial settings, integration in production line and testing.
- Bring new insights including industrial perspectives on identifying and quantifying the benefit from using undisclosed high valuable sources in order to improve modeling and calibration performance in the context of data modeling for industrial and mobility applications, in line with the **Industry 4.0** movement (see Fig. 2) and the **Internet of Things** using mobile and RFID technologies and data fusion from different sensors, where appropriate.
- **Use of an algorithm which is a patent application co-authored by Dr. Nettleton** (See Ref. 3, Sec. 1.3.1) in comparison with other considered state of the art data mining methods. This algorithm consists of a matching technique for data represented in graph like structures, such as is the case in a wide spectrum of domains such as online social networks, recommender systems, urban transport planning, logistics and telecommunications and molecular design/drug discovery, among others. As part of the work, a front end will be developed for the algorithm as a generic demonstrator system. Then for a chosen case study, the algorithm will be customized for the given data, matching requirement and business objective. The specific case studied to which it will be applied is expected to be the urban transport project, however other applications can be chosen as alternatives, if necessary.

For the computerization of manufacturing, focussed and market-driven projects such as DATAMINE4.0 are necessary for ensuring European excellence in the face of continuous global innovations and advancement of the worldwide industrial manufacturing technologies, which is typically regarded as a key area of competitiveness. One of the specific objectives of the H2020 industrial thematic area is to boost European leadership in the manufacturing sector as KETs (Key Enabling Technologies). Industry 4.0 refers to an industrial revolution based on cyber-physical systems, which has its origins as a German Industry initiative. In short, the technical challenge in this project of performing data analysis and modeling to solve real industrial problems and improve existing processes, will make it possible to perform processes (such as in-line customized mass production) which were previously impracticable. This is facilitated by the availability of machine learning and statistical function libraries which can be integrated into industrial systems, in closed loop or outside as an initial calibration step. Data analysis and modeling of complex production machinery and multiple step in-line processes in pre-industrial scales and that achieves customization competing with low volume production iis highly innovative. Information on IP is given in part 3.2.3.

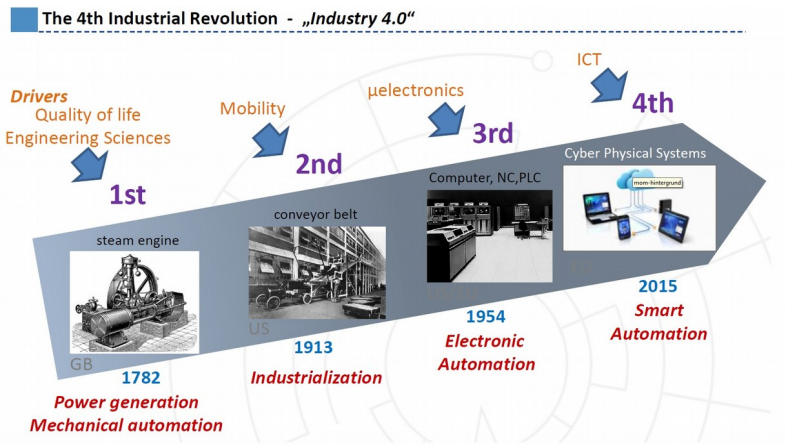

Fig. 2 Evolution towards the fourth industrial revolution: Industry 4.0

1.2. Quality and appropriateness of the training and of the two way transfer of knowledge between the researcher and the host

During her PhD studies, Natasa has performed modeling of telecomunications network applications at the Ecole Polytechnique Fédérale de Lausanne, Switzerland, thus obtaining both academic and industrial know-how and experience in applying advanced data modeling techniques to real world problems. During that time she gained experience and insight in the development and application of statistical models applied to the testing and development of human mobility, network routing and simulation. The DataMine4.0 project will allow her to widen the application areas of her statistical analysis skills to other domains, as well as potentiating the projects using novel algorithms. Also, it will be an opportunity to augment her curriculum in publications in congresses and journals afin to the DataMine4.0 objectives.

Working in a multidisciplinary network, Natasa will increase and enrich her scientific knowledge in the analysis, development and deployment of industrial data modeling. She will also have an unprecedented opportunity to deploy powerful data mining solutions in real industrial production environments and thus contribute to the Industry 4.0 and Internet of Things movements. She will be trained on new and powerful research and application techniques, transferring from a test environment to a pilot industrial scenario under the guidance of data modeling scientists with experience in industrial data modeling validation and testing, with a team of engineers who will help him to transfer the results to the industrial requirements while giving her exposure to many transversal skills. On the other hand, we can also learn from Natasa's specialist knowledge of telecommunications network modeling, which in the future will be increasingly a strategic component of IRIS's industrial applied research projects, making use of mobile and radio frequency identification in new scenarios.

Furthermore, as in IRIS there is very dynamic environment and given the fact that there are numerous projects with academic institutions, Dr. Sarafijanovic-Djukic will cooperate and learn from some of the best scientists in the field (see section 2.3). Also, she will spend 2 months each at the University of Leuven - KU Leuven (Belgium) and the University of Skövde (Sweden), where she will attend courses and hands-on collaboration in order to obtain knowledge on industrial and applied graph matching and data fusion, respectively. Both of these themes are key data sciences hot topics and are strongly related to the DATAMINE4.0

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proposal: graph matching and data fusion to obtaining a strong set of inputs for the data models. Likewise, she will directly apply learned knowledge in the evaluation and development of data mining projects. In addition, Natasa will enrich his curriculum with new skills in innovation management, technology transfer to the industry, project and IP/knowledge management, as well as extend her article writing and presentation skills for conferences and journals.

Scientific training objectives (SCO) will be fulfilled as a result of the training-though-research under supervision and hands-on training, through this individual personalised project that has been agreed between the Fellow and the Host (Table 1). Note that WP2, 3 and 4 contemplate a number of case studies. Audits will typically have a duration of 1-3 weeks and the projects from 2-6 months. Therefore it is contemplated to complete 3 or 4 case study projects (among those previously listed) during the 2 year period.

WP1- Literature background and definition of specifications
SCO - Training on specification definitions for market driven and applied research based on a mapping of the existing research and commercial state of the art against market requirements for new solutions. Quality Function Deployment.
Training verification (TV): Presentation of the specifications to the Innovation Committee (IC) and Scientific supervisor and subsequent approval.
WP2- Data Audit – requirements studies
SCO-1) Improved skills in evaluation and requirements capture of data mining projects using data audit methodology; 2) Incremental development approach based on Evaluation of Business Objectives & identification of necessary and available data. A requirements report is produced for each case study.
TV: Data Audit report for different case studies presented to IC and Scientific supervisor and subsequently approved.
WP3- Data Project – execution of projects
SCO - Execution of projects defined by WP2 (Data Audit) using data miodeling and statistical skills in different data and industry domains; 1) Data acquisition and pre-processing; 2) Data Analysis; 3) Data Modeling; 4) Testing and verification.
TV: Development of data models in new domains as a new learned skill for the Fellow that will be transferred to the Host. The evaluation of model precision and improvement on existing methods will allow verifying good practice.
WP4- Transfer of knowledge and technology to industrial scale, industrial validation and production
SCO – Deployment. 1) Integration in Industrial production processes; 2) Transfer of technology
TV: Successful integration of data models in the industry and its positive comparison with current methods leading to the engagement of the industry to further uptake this new solution.

Training activities will be scheduled following Table 1 in order to provide **structured training** needed by the fellow.

Table 1. Programme of training courses to be delivered to the Fellow

No	Type of course	Short description of course and expected impact for Fellow	Date
1	Project Management (PM) Training course	Training on ISO certified PM methodologies & procedures, including latest in PM Institute principals and vanguard lean PM methodologies (e.g. High Level One Page Project Planner method, Sapenta), and Risk Management procedures that provide skills in improved time, resource & quality management to achieve project objectives.	M1
2	Seminar on communication with industry	Seminar on effective communication towards decision makers in industry, in order to provide the Fellow with new skills that will be beneficial when engaging with industry - Improved scientific professionalism through the publication of findings in leading sectorial/industry journals.	M4
3	IPR Seminar	Seminar/patent attorney or IP expert - Innovation Management skills	M7
4	Practical "data audit" studies of real companies	Training delivered by the Host to transfer knowledge to the Fellow- Learning of techniques to characterise standardized properties of interest when executing a data mining project consisting of analysis, modeling and calibration phases. Training which includes data aggregation/fusion delivered to the Fellow in collaboration with the University of Skövde (1st SECONDMENT), months 9 & 10.	M8/ M9-10
5	Practical "data project" data mining of real companies	Training delivered to the Fellow in collaboration with the University of Leuven - KU Leuven- Learning of techniques to characterise standardized *modus operandi* when executing a data mining project consisting of analysis, modeling and calibration phases in an industrial domain. As part of this training, the adaptation of graph matching algorithm (patent) to a real application (2nd SECONDMENT), months 17 & 18.	M9/ M17-18
6	Workshop on data modeling at IRIS	Overview of the key results and skills of data modeling, analysis and evaluation metrics at IRIS, which will improve the Fellow's ability to document and present scientific results to a wide audience	M21
7	Training through organisation of demo sessions to industry	With assistance and guidance from the Host, the Fellow will improve her organisational skills through the organisation of a demo event to demonstrate the project results to industry, and to give the Fellow exposure to high level industry decision makers. Industry exposure and an applied, industry-driven research focus	M24

1.3. Quality of the supervision and of the integration in the team/institution

1.3.1. Qualifications of the supervisor

The researcher in charge is Dr. David F. Nettleton, the head of the Data Science Group in IRIS who has extensive experience and know-how acquired over more than 20 years in data mining and statistics projects in Industry as well as applied research in different Universities. His work has ranged from web user behaviour modeling and analysis, to data privacy analysis, social network analysis, medical diagnosis and prognosis, among others. Dr. Nettleton has led and participated in R&D projects aimed at building novel data models and data processing methods, matching methods, developing new analytical methods and modelling; as well as collaborating with academic labs and other players in the IT industry (including working for 4 years in IBM Global Services as Business Intelligence Consultant). He has been thesis director of two Ph.D. students, one Masters and four undergraduate final year projects and has 8 years teaching experience in 3 Universities. Since joining IRIS in 2015, he has coordinated or been involved in the technical management or scientific contribution in several EU research projects under H2020 (OpenMind, SmartGran, Sharebox). This makes David an experienced adviser on data science and its application to provide solutions in real scenarios which are crucial for this research project. Among those most related with the current proposal, David is Technical Manager of the OPENMIND project (On-demand production of entirely customized minimally invasive medical devices, GA 680820); has contributed to the SHAREBOX project (Secure Management Platform for Shared Process Resources, GA 680843) and SMART-GRAN (Development and Commercialisation of a Self-Guided Fluidised Bed Granulation Solution, GA 666588).

These projects have afforded IRIS the opportunity to forge active and lasting international collaborations with International institutes and Universities renown in the field, such as Fraunhofer Institute for Production Technology IPT (Germany), Katholieke Universiteit Leuven (Belgium), Universitat Politécnica de Catalunya (Spain), Universitat Pompeu Fabra (Spain), IIIA-

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CSIC (Spain) as well as numerous industrial enterprises such as Diribet (Czech Republic), Gimacs (Italy), In-Core Systems (Netherlands) and Inno Pharma (Ireland), among others. Dissemination and outreach to communication media has attracted significant industry attention. David will transfer invaluable expertise regarding knowledge in data science and data model development for industrial applications. In addition to his publication track record, he is author of three complete books, numerous papers in peer-reviewed and technical journals, Open Access repositories such as Academia, Researchgate and Arxiv, and has presented work in many international conferences. Dr. Nettleton has over 70 publications (many avialable in ResearchGate, Academia and Mendeley) with 8 SCI journals, 3 full books as principal author and over 40 congresses. Some relevant publications are listed below while additional publications from the team are listed in Section 6:

1. Nettleton, D.F., Commercial Data Mining - Processing, Analysis and Modeling for Predictive Analytics Projects, 1st Edition (2014), Morgan Kaufmann Publishers, ISBN :9780124166028, 304 pages. **2. Nettleton, D.F.**, Orriols-Puig, A., Fornells, A., A study of the effect of different types of noise on the precision of supervised learning techniques. Artificial Intelligence Review, April 2010, Volume 33, Issue 4, pp 275-306. **3.** Local Neighbourhood Sub-graph Matching Method, International and European Patent Aplication PCT/ES2014/065505, Inventors: **David F. Nettleton,** Anton Dries. **4. Nettleton, D.F.**, Bugnicourt, E., Wasiak, C., Rosales, A. Towards Automatic Calibration of In-Line Machine Processes, 18th International Conference on Industrial Engineering and Manufacturing Process Modeling, ICIEMPM 2016, London, United Kingdom, July 28-29, 2016.

1.3.2. Hosting arrangements

IRIS will provide all required administrative and logistic support for Natasa's settling. The Barcelona area is a perfect place for relocation of international families like Natasa's as it has all infrastructures and education systems necessary to host non-(yet) Spanish speaking people (including even a Swiss school, http://www.escuelasuizabcn.es/es/). As such, Natasa, helped by her family mobility allowance, is convinced that the Marie Curie fellowship can not only be an excellent opportunity for herself but also a successful experience for her family.

Our candidate has been conciliating her motherhood (2 kids born in 2011 and 2014 respectively) and part time freelancing during 4 years while also taking the occasion to up-skill by different training regarding statistics. IRIS, which is immersed in a gender equity culture, and in fact co-founded and directed by a woman, is convinced that such strength can help Natasa to achieve the goals of the proposal and that giving the chance to women to a fair reinsertion to work after a break linked to the birth of a child is something that can only help our society grow towards tomorrow's standards in line with improving the, too often lacking, gender equity spirit in businesses.

1.4. Capacity of the researcher to reach or re-enforce a position of professional maturity/independence

Natasa has a strong track record in data modeling applied to telecomunications networks and mobility. She obtained Diploma of Electrical Engineer in Telecommunications at University of Belgrade in Serbia, coming in the **top 3% of the promotion**. In these studies she learnt about different stochastic models of telecomunications systems. This was followed later by her PhD studies in Computer/Communication Sciences at EPFL, Switzerland, in which Natasa applied data modeling to telecomunications networks and mobility. Her initiative to study abroad, moving from Serbia to Switzerland, shows her independence and ambition for an **international projection**. Natasa's competiveness and professionalism is shown in her drive to gain both deep theoretical knowledge and a wide-spectrum of practical skills in order to successfully apply them in real-world problems. Also she successfully realised her Ph.D studies in combination with starting a family.

With such background, Natasa appears to have the capacity to contribute to DataMine4.0 success which will in turn **maximize her potential in applying fundamental knowledge in data mining to industrial and urban planning processes**. The mobility afforded by this Fellowship will be a new enriching challenge in Natasa's career. Such **international** industry-focussed work experience along with the know-how and strong networking opportunity **will be invaluable sources of knowledge** to enhance Natasa's attractiveness as an excellent researcher in the ERA and even beyond. Working within a research organisation, dialoguing and engaging with academics and researchers, establishing research plans, performing experiments in order to evaluate and publish results, are all aspects that Natasa is highly familiar with and proficient in. However, industry needs and motivations are very different to those of the academic community, and indeed the priorities and pace of industry are often quite different from those of academia. In the context of an Innovation Union, the whole concept of scalability needs to be at the forefront of a researchers mind when operating in the applied space, whereby the ability to eventually fulfil large-scale production requirements is paramount for industrial success. **The gaps in the academic research provide opportunities** (see 2.2.) for Natasa **to develop her authorship skills in publishing articles in congresses and journals, as well as project management and coordination skills** in order to **reach the next level as a researcher** which will facilitate her in achieving also a position of professional maturity, diversity and independence. The research experience proposed within DataMine4.0 will be the next logical step to complete Natasa's training and is an essential step in her career since: 1) Natasa's career goal is to perform **applied science and lead her own project;** 2) it provides **multidisciplinary and intersectorial scientific training** (including food processing modeling, urban mobility optimization, calibration of customized industrial production) that will be a complementary exceptional experience. **This will also make use of her telecommunications expertise in choosing applications which embrace the "Internet of Things" paradigm, requiring, for example, data fusion from heterogeneous mobile, RFID and other sensor devices;** 3) Natasa will **benefit from dynamic and proactive exchange of knowledge** in formulation, processing and **implementation of research findings in order to reach scientific excellence and be able to expand his network for her future career path.** This will be potentiated by two secondments with the research groups of two distinct universities with strong track records in theoretical and applied data science. The engagement will help in strengthening initiative and polyvalent skills providing freedom in independent thinking and project management, while still working as a part of dynamic team with experts in the field (in company or through the external academic collaborations described in section 2.2). Finally, currently speaking two European languages (English and German), learning Spanish during this Fellowship will **make Natasa even more marketable in international business**. In short, this fellowship will be a key step to lead Natasa towards the career she wants to embrace in the forthcoming years in IRIS and beyond.

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2. IMPACT

2.1. Enhancing the potential and future career prospects of the researcher

Researchers usually make a choice to either work in academia or in industry. The telecommunications field is somewhat hybrid in this sense, given that it requires having a presence in both ambits, which is the case of Dr. Sarafijanovic-Djukic and results in acquiring a **dual-competence**. As was seen, the scalability of results needs to be at the forefront of a researchers mind when operating in the applied space, whereby the ability to eventually fulfil large-scale production requirements is critical for industrial success. This 2 year fellowship in IRIS, a private company, provides the perfect bridge between scientific excellence to the transversal and applicative skills that complement the academic environment. These skills will further equip her to become an excellent, well-rounded researcher who is even better prepared to participate in the **European Innovation Union and its "internal market" for research and transfer scientific knowledge towards industrial applications.** Attending meetings with the research and engineering teams, experiencing the day-to-day procedures in IRIS, being a member of the Innovation Unit, are **invaluable opportunities for gaining an insight of the strategic decision-making and management process within a commercial applied research/advanced engineering environment**. There is a strong innovation dimension of this Fellowship, which is in line with her entrepreneurial spirit, which has the ambition to boost R&D in order to generate business, and to strengthen industrial leadership in innovation by helping to bridge the gap between research and industry. Identifying, protecting, and exploiting valuable IP unite highly powerful strengths, especially telecommunications know-how applied to projects within the “Internet of Things”. Natasa’s scientific excellence and applicative science will be linked and no further gaps will exist at the current stage of her career. Natasa is very confident that this Fellowship will contribute abundantly to the development of his career in the medium and long-term, given that he will be involved in **the leading edge of the interdisciplinary applied research**, **with thorough training in applying academic concepts in practice**, and to the opportunity to explore possible future occupations. Indeed, this is timely, as the implementation of science to serve industry innovation is the main instrument proposed by Horizon 2020. Natasa embraces this ethos and wants to be a major research contributor to the success of this strategy. As a perfect place to launch herself on that trajectory, IRIS is a company that has strategically nestled itself between academia and industry, combining a scientific with an advanced engineering expertise to transition knowledge and technology from the lab right through to production lines in the industry.

2.2. Quality of the proposed measures to exploit and disseminate the action results

2.2.1. Dissemination of the research results

The expected results of this project will lead to a significant increase in European excellence in the field of applied data science, especially data modeling and calibration using a diversity of statistical and machine learning techniques applied to Industry 4.0 applications in the context of mobilising European knowledge to take the lead in the Industry 4.0 movement, as well as ambits which can improve the daily life for millions of EU citizens, such as urban mobility planning. In addition to the outreach activities in Table 2, additional effort will be made in terms of outreach to industry. The applicability to industrial and public local organizations of the DATAMINE4.0 work is expected to greatly benefit the EU industrial and public urban planning sectors and as a result of increasing competitiveness in the region and penetration in the international market, industrial profitability and improved life quality for citizens. Moreover, as the project offers case studies in different ambits, initially defined by, but not limited to the chosen case studies of (a) food processing modeling, (b) urban mobility modeling and (c) customized production of composite materials components, the knowledge generated during the execution of DATAMINE4.0 will be disseminated in diverse areas of applications such as the ubication of pedestrian crossing and bus stops, medical composite device manufacture, diagnosis and solution of problems arising during industrial food processing. **These projects will embrace both the Industry 4.0 and Internet of Things paradigms, which will be hot topics in forthcoming years.** The dissemination aspect is important in the case of Natasa she has taken a break of several years due to starting a family, although she has continued to take new training courses and more recently as freelance Data Scientist. Thus the fellowship is a great opportunity for Natasa to potentiate her publications and her career, thus, although keeping an eye on the IP protection and confidentiality of final detailed models, in IRIS and at the secondment institutions we will take special care to make sure that a fair part of the work can be disclosed in the public domain, targeting quality congresses (ACM / IEEE) and journals (SCI) for submission of her work. The supervising researchers, especially Dr. Nettleton (IRIS), Prof. Torra (Univ. Skövde) and Prof. Jesse Davis (ML Research Group,KU Leuven) have strong track records in publications in the Data Science field and are currently very active in journal and congress submissions. Another key aspect of dissemination will be the guarantee of Open Access to publications. This can be done by additional payment to Journals, but also can be achieved by choosing green open access publications, by publishing pre-print versions in online repositories, preferably indexed in scientifically recognized search engines like Scopus, such as Academia, Researchgate and Arxiv, among others.

2.2.2. Exploitation of results and intellectual property

Once the public results are listed, the good management of the confidential IPR will maximise the exploitation potential as further discussed in 3.3.4. In a research scenario where industry is an active collaborator, patentability is an important factor to be considered. A basic patent review show *a priori* freedom to operate. This search will be deepen in the project building on IP skills gained by the fellow. The Results generated will be defined as they emerge considering issues such as novelty, patentability and protectability. A review and analysis of the most appropriate IPR tools to protect the results will be carried out and patent (or other IPR) applications will be filed for. Trade/Industrial secret and know-how could also be powerful IP tools.

2.3. Quality of the proposed measures to communicate the action activities to different target audiences

In keeping with the public engagement principles of the 'European Charter for Researchers' and 'Code of Conduct for the Recruitment of Researchers', this project will ensure that the research activities are presented to society. In relation to the benefits for the Fellow of performing these outreach activities, direct engagement with the public will help her to better understand public interest in priorities for science and technology and also the public's concerns. In addition to peer reviewed papers to be delivered in the research Work Packages (WP), a comprehensive Outreach Activities Plan is presented in Table 2.

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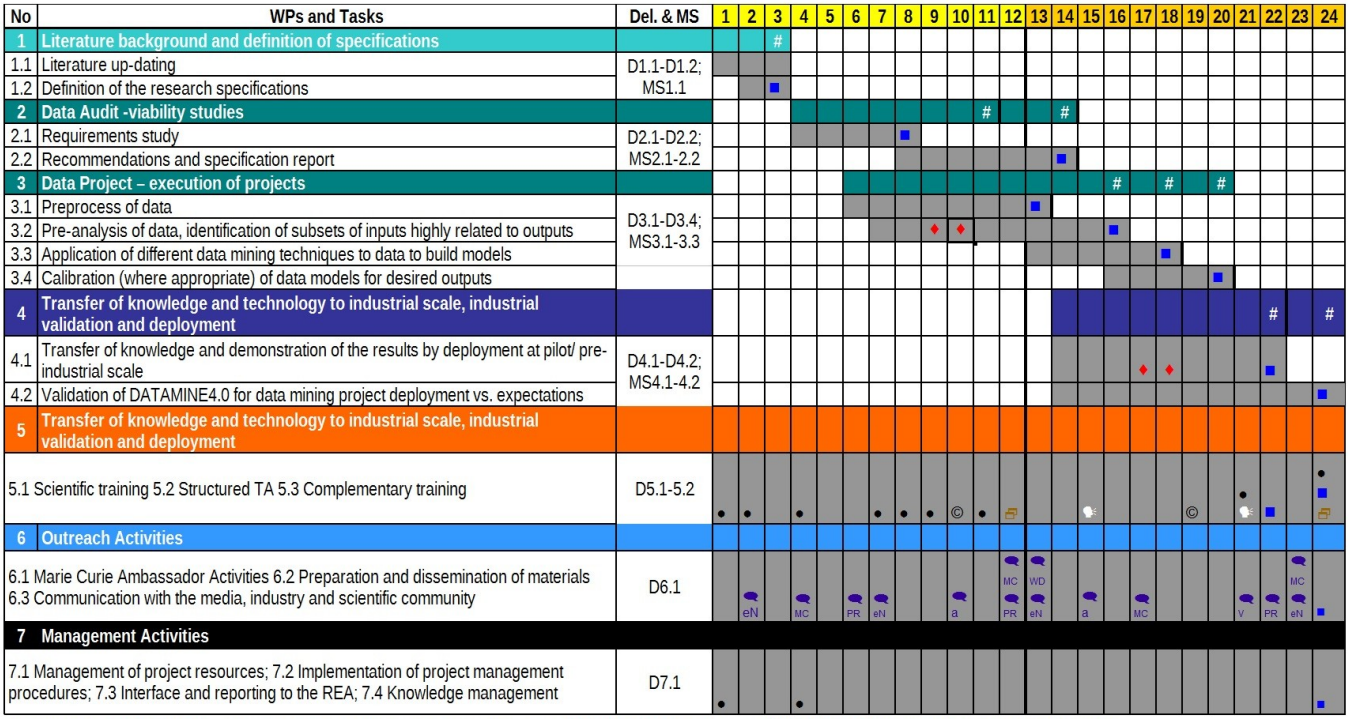
Table 2. Plan of Outreach Activities

Activity Type	Description	Target	KPIs	Expected impact
MC Ambassador actions	Visits to schools of the Polytechnic University of Catalonia (UPC) and secondary schools in the local community to promote research field and about benefits and rewards of being a researcher	University students, secondary students	- Minimum of 4 visits to be made over 2 year project - Over 150 students reached	To attract more young people to take up scientific careers and to promote Europe as an attractive place to study and perform research.
Workshop Day	*"A future in Science"* workshop/activity day for school/uni students at the IRIS facilities/ the campus of the UPC in areas related to the raising of scientific awareness (to be organised in conjunction with the Mayor's Office)	University students & secondary students	- 1 day activity (half day seminar, demonstration session, tour of facilities, career guidance sessions) - > 150 students reached	
Press releases (PR)	Written communication about the project that will be directed at members of the news media and edited to ensure the purpose of announcing something ostensibly newsworthy.	General public	- 2-3 PR issued at key points - 5 regional or national newspapers & media channels	Increase awareness among the general public of the importance of R&D investments and how scientific achievements can have a direct and positive impact on people's lives.
e-Newsletters	A web-based document to be released on the internet for the attention of the public at large (e.g. Wikipedia).	General public	- 4 newsletters published - 300-500 downloads of each newsletter	
Articles in newspapers/ interviews in TV	Through collaborations with the Mayor's Office and using the Marketing Dept. in IRIS, feature articles and/or local TV interviews will be organised with the Fellow	General public	- At least 2 press articles or TV interview	
Marie Curie project video	Video clip about the project, the Fellow and the MC experience on YouTube, IRIS website...	General public	- Over 1000 views	

3. QUALITY AND EFFICIENCY OF THE IMPLEMENTATION

3.1. Coherence and effectiveness of the work plan

In the following we detail the Work Packages, Tasks, Milestones and Deliverables. Four core Work Packages are designed to lead the researcher through a logical sequence of phases, each of which builds on the previous one. WP1 collects and organizes the necessary background knowledge in the context of DATAMINE4.0, focused on applied research, but also containing novel approaches which will add to the current state of the art. As we described in Section 1.1, a methodology consisting of two major phases will be applied to design and execute different data mining case studies. This is reflected in WP2 (data audits/requirement studies) and WP3 (data projects). These are followed by WP4 which contemplates the deployment phases in which successful projects are implanted in the business/organization of the collaborating entity. Finally, Training, Outreach and Management activities are described in their own respective work packages (Wps 5, 6 and 7, respectively).



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WP1- Literature background and definition of specifications

Start Date	M1	**End Date**	M3

Objectives ■To position the research activities in line with the state of the art, hot topics and market requirements■To identify and confirm current approaches, methodologies and tools for an applied research context of intelligent data modeling/calibration, statistical analysis, machine learning and artificial intelligence techniques, knowledge and data representation, as well as contempory algorithms, software packages and approaches for data mining. **All of this will be performed with an emphasis on the Industry 4.0 paradigm.**

Task 1.1. Literature up-dating. Algorithms, techniques and methodologies to be used. Relevant up-to-date "best of breed" case studies with cost-benefit analysis to identify high potential projects.

Task 1.2 Definition of the research specifications Specific market requirements for the implementation of new applications for intelligent data modeling in terms of technical feasibility and within the Industry 4.0 context. Identification of sectorial solutions, definition of process specifications, critical parameters. Commercial viability and market potential of applications and their cost/benefit. Metrics for quality control and evaluation of results and impact.

Deliverable(s): D1.1 Document/Report of the state of the art and literature review (M3); D1.2 Document/Report of the research specifications (M3)

MLs & KPIs ML1.1 Specification report and its approval (by IRIS IC) as a reference document for the R&D effort for the remainder of the work plan (M3)

WP2- Data Audit -viability studies

Start Date	M4	**End Date**	M14

Objective ■ To carry out requirements studies of different scenarios in industry/commerce/business/public entities, applying a systematic methodology that fulfils specifications defined in WP1. ■ Visits, interviews and meetings with stakeholders in organizations/companies in order to elicit information about project objectives, data required/available and necessary resources. ■ To quantify the evaluation using a systematic methodology to calculate cost/benefit for each potential project. ■ Based on the quantified evaluation of cost / benefit, choose the top most viable business objectives, preparing a corresponding report with a detailed justification for each.■ Three generic application areas for the case studies will be (but not limited to): (a) food processing modeling, (b) urban mobility modeling and (c) customized production of fibre reinforced plastic components. Others may be defined later.

Task 2.1 Requirements study. For each case study, follow a systematic methodology to elicit key information: business objectives, available data, data quality and availability, inputs and outputs.

Task. 2.2 Recommendations and specification report. Based on the requirements study, recommendations are made in terms of necessary data and execution plan for the project. The defined case studies will be used for next WP.

Deliverable(s) D2.1 Report on the requirements for each data mining project (M8); D2.2 Recommendations and specification report (M14)

MLs &KPIs ML2.1 Identify at least 1 project from each case study main area (M11); ML2.2 All top viable projects identified and requirements defined (M14)

WP3- Data Project – execution of projects

Start Date	M**6**	**End Date**	M20

Objective ■ To execute the projects defined in WP2 ■ To build data models with the best set of inputs and ouputs in each case ■ To calibrate data models (where appropriate)■ To characterise processes automated by data models in comparison with previous/current methods ■ To identify data models which can subsequently be deployed in WP4

WP3 will be run in parallel with feedback from WP2. (Note that in some cases, given aspects of the data audit in WP2 are re-executed, for example, in order to obtain new data / data values)

Task 3.1 Pre-process of data. For each viable case study, data identified in WP2 is obtained (extracted from proprietary databases, etc.) and preprocessed into an adequate form for data modeling. This includes data cleaning and consolidation, identification and processing of erroneous and missing data.

Task 3.2 Pre-analysis of data, identification of subsets of inputs highly related to outputs. For each viable case study, once adequate datasets are available from the previous Task 3.1, a pre-analysis can be conducted to select inputs and outputs for data modeling. This includes sampling, derivation of new factors, variable selection using techniques such as PCA.

Task 3.3 Application of different data mining techniques to data to build models. Depending on the type of data modeling (prediction, classification, explanatory, time series, etc.) different algorithms and paradigms will be tested to build models. The algorithms which give the most consistently high precision will be chosen.

Task 3.4 *Calibration (where appropriate) of data models for desired outputs*. This step will be applied where calibration is necessary (e.g. to find optimal inputs to produce a given output and thus assign these values to, for example a PLC for customized manufacturing processes).

Deliverable(s) D3.1 Datasets prepared in an adequate form for analysis (M13); D3.2 Detailed report on data pre-analysis for each case study (M16); D3.3 Data models for each case study (M18); D3.4 Calibrated data models (where appropriate) (M20)

MLs&KPIs ML3.1 Dataset preparation and pre-analysis done (M16); ML3.2 Data models implemented (M18); ML3.3 Data models calibrated (M20)

WP4- Transfer of knowledge and technology to industrial scale, industrial validation and deployment

Start Date	M**14**	**End Date**	M24

Objective ■ To deploy application of the selected data models of selected case studies in pilot scale facilities ■ To evaluate the results and improvement versus expectations for calibrative/classificative/predictive data models ■ To draw up recommendations for full scale industrial deployment of data models on production /real environment and evaluation with their real applied needs

WP4 will be start as soon as availability of data models is granted for the respective case studies from WP3.

Task 4.1 Transfer of knowledge and demonstration of the results by deployment at pilot/ pre-industrial scale. The scaled up deployment of the best data project models developed in WP3 will be carried out for each pilot installation of the selected collaborating SME/Organization.

Task 4.2 Validation of DATAMINE4.0 for data mining project deployment vs. expectations. The best data model deployments obtained in pilot/ semi- industrial scale will be used for validation tests with support from the Fellow to demonstrate their feasibility to replace or optimize currently used methods

Deliverable(s) D4.1 Report on deployment of data models produced at semi-industrial scale (M22); D4.2 Report on the use of DATAMINE4.0 validation for data mining project development and deployment (M24)

Mls & KPIs ML4.1 The selected project data models are successfully deployed in the pilot locations (M22); ML4.2 DATAMINE4.0 is recognised as suitable approach by its deployed site stakeholder (M24)

WP5- Training Activities

Start Date	M1	**End Date**	M24

Task(s): Task 5.1 Scientific training. Task 5.2 Structured TA. Task 5.3 Complementary training. *See sections 2.2. & 4.2.3*.

Deliverable(s) D5.1 Report on scientific knowledge transferred from the Fellow to the Host (M22)
D5.2 Report on knowledge transfer sessions delivered and materials (M24)

WP6- Outreach Activities

Start Date	M1	**End Date**	M24

Task(s) Task 6.1 Marie Curie Ambassador Activities. Task 6.2 Preparation and dissemination of materials. Task 6.3 Communication with the media, industry and scientific community. *See section 3.2.1.*

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Deliverable(s) D6.1 Public report on the project outreach activities (M24)			
WP7- Management Activities			
Start Date	M1	**End Date**	M24
Objective ■ To perform structured and coherent Management activities linking together all the project components and maintaining communication with the Research Executive Agency (REA). ■ To manage the knowledge/results generated during the project for heightened impact.			
Task(s) Task 7.1 Management of project resources. Task 7.2 Implementation of project management procedures. Task 7.3 Interface and reporting to the REA Task 7.4 Knowledge management. *See sections 3.2.2 and 4.2.4*			
Deliverable(s) Contractual reporting. D7.1 Register of IP outputs and protection routes (M24)			

3.2. Appropriateness of the allocation of tasks and resources

The work plan has been drawn up jointly by the Fellow and IRIS and both parties are confident that it is achievable and realistic. The objectives have been used as the starting point for drawing up an appropriate work plan, with dedicated WPs, broken down into tasks, with clear outputs as deliverables. This plan is considered realistic and achievable by both parties for effective implementation within the competitive 2 year timeframe, and offering a good level of ambition to ensure the Fellow benefits from the research work for improving her skills and capacity. The objectives and tasks involved in each research WP are described in Section 3.1, along with the expected deliverables and clear Milestones (MLs), Key Performance Indicators (KPIs) that will help assess the progress of the project (note the training that the Fellow will benefit from as a result of the implementation of each WP is discussed in 1.2.). For the other WPs, outreach and training tasks are briefly presented and are further described in management activities (section 3.3).

3.3. Appropriateness of the management structure and procedures, including quality management & risk management

3.3.1. Project organisation and management structure

Our practical and effective management structure is based on a simple, yet effective structure (Fig. 3), whereby the Fellow acts as the Principal Investigator and Project Manager for the internal research team. She will report directly to the Data Science Group Leader, who will provide supervision and support to the Fellow in the management and implementation of the work programme and the project team. Both will form part of the Innovation Unit body in IRIS that reports on a quarterly basis to the Innovation Committee (IC). Day-to-day engagement between the Fellow and the Project Team in IRIS, inner academic communications such as group meetings, design discussions, presentations will ensure informal yet highly effective and continuous knowledge transfer from the multidisciplinary team to the Fellow. The Fellow will also benefit from the support and expertise on project management principles, as well as financial and administrative management of European projects, training on processes and procedures followed in IRIS in keeping with their ISO 9001:2008 and UNE-EN-ISO 166002: Management Systems for R&D&i approved processes. The work plan, which includes the objectives and milestones that will facilitate the management of the project and the assessment of its progress against what was planned.

3.3.2. Research and/or administrative risks that might endanger reaching the action objectives

The process of risk identification and management will be based upon the principles and guidelines of the FERMA (ISO) Standard. The risk management process will be based on a list of coordinated activities embracing the 7Rs and 4Ts of (hazard) risk management. A list of a priori risks has been identified and is listed below along with the contingency plan in each case.

WP2	R1.1-One or more of the initial projects/objectives studied are inviable. CP1.1-This risk is low as the list is open and consists of initial case studies suggestions with number of sub-options but a backup list of potential projects and alternative business objectives could be defined.
WP3	R2.1-Requirements study incorrect, unforeseen problems in carrying out project. CP2.1-The requirements studies will be rigorous and realistic and only if absolutely necessary, we could reduce expectations and/or adapt/revise objectives (affecting a specific case study and not all). Typical problems at this stage are to do with data quality, objectives, data availability. Dr. Nettleton has a wide experience of mitigating these types of problems in real projects.
WP4	R3.1- Models and calibration work well in test environment but not in production. CP3.1- Start validation work early enough to allow iterations with previous WPs. Identify what are the differences between the test and production environments and make adjustments to models and calibration, try different paradigms (families of algorithms), and algorithms within the same paradigm. R3.2 Patent algorithm (graph matcher) does not work or cannot be implemented. CP3.2 – Although this approach is only one of the considered, limiting the impact of the risk, it is expected to be most suitable for urban case study. The graph matcher has already passed a proof of concept test with a commercial company (Nubelo online employment web) and has been adapted to different uses (such as social network search and data privacy). Dr. Nettleton is co-inventor, has the source code and in depth knowledge of how to adapt the algorithm to new data and contexts.
All WP	RN.1- The candidate decides to prematurely leave the project due to work profile not corresponding to expected one and/or due to lack of matching of her skills with the needed one. CPN.1 The selection procedure has been rigorous and has taken into account this risk during selection whereby Natasa appeared to best fit the project needs among the other candidates shortlisted. After discussing the proposal through several meetings, the candidate has agreed to, and indeed improved, the proposed work. The secondment placements are also designed to augment the skills of the candidate, as well as the transfer of know-how and support by IRIS through the supervision of Dr. Nettleton. RN.2- The candidate decides to prematurely leave the project for personal reasons. CPN.2 We have a personal commitment from the candidate and a concrete plan for taking care of her family arrangements and requirements (see Section 1.3.2). RN.3- One or more of the secondment institutions fails to fulfil one or more of the secondment requirements (including pulling out completely) or lack of supervision. CPN.3 The secondment institutions, with whom there is previous relation and confidence, has carefully chosen. Also, both institutions have the necessary know-how in Data Science and multiple persons who could be supervisor. Finally, we have two secondment institutions so we expect a worst case scenario of one failing having the other as backup in which case we could have the option of extending the stay, and/or establishing an agreement with another University and/or organizing tutoring in remote without forgetting that IRIS already had most of the core required knowledge to train Natasa and supervise the work, including some knowledge of graph matching and data fusion. RN.4 Failure of IRIS to host the researcher. CPN.4 IRIS has a strong track record of supervision of undergraduate, Masters, Ph.D and post-doc supervision, including one current Marie Curie Research Fellow in the Materials Science field. Also, IRIS is currently active partner in many H2020 projects with pipeline of work in hand reaching 3-4 years ahead, many new proposals being prepared and has its own lines of industrial monitoring hardware, thus it represents a stable partner for applied science projects and training (see Section 1.3.1). Also, this proposal represents a strategic area for IRIS in order to potentiate its offering of services so the likelihood of the host losing interest in the project is null.

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3.3.3. Progress monitoring and project management procedures

In relation to the **project management procedures** that will be followed to ensure that the project is successfully managed, IRIS operates to proven standard project management and ISO 9001 principals, founded on sound project planning and subsequent monitoring and controlling of the execution of the workplan. These project management procedures consist of: Technical Workplan Review and Updating; Issues Management; Scope Management; Risk Management; Conflict Management; Communication Management; Document Management; and **Quality Management (using the QFD tool)**. IRIS has edited a comprehensive handbook of project management procedures with supporting tools and documentation, which will be clearly explained to the Fellow through induction training. Following the training and induction period (1-2 weeks), the project will officially start with the Kick-Off meeting that will be chaired by the Fellow and in which the Project Scientific Supervisor and the Project Team will participate. Afterwards, the project will adhere to the following meeting and reporting schedule:

Meeting type	Frequency	Who attends	Associated Reporting
Project Team Meeting	Weekly	- Project Manager/PI (Fellow) - Project team	Weekly report from team to the Project Manager (using standard template)
Scientific meeting	Fortnightly	- Project Manager/PI (Fellow) - Data Science Group Leader (Scientific Supervisor)	Project Manager reports progress of the work to the Scientific Supervisor (using One Page Project Planner tool in use in IRIS)
Innovation meeting	Monthly to IU, quarterly to IC	- Project Manager/PI (Fellow) & Scientific Supervisor (as part of the Innovation Unit IU) - Innovation Committee IC	Report to the IU on a monthly basis and to the IC on a quarterly basis on the status of the project and its IP outputs as well as the possible applications of the knowledge and results generated to other fields.

All meetings require an agenda circulated to attendees where record key statements, recommendations and actions (who, what and when) will be documented. Project will also adhere to the reporting and review meeting requirements of the Grant Agreement with the REA. The IRIS intranet will be used as a tool for facilitating the project management effort, whereby the Project Gantt (presented above) will be dynamically managed in a "One Page Project Planner" version for enabling High Level Project Management. IRIS follows an agile development methodology for the operational development of tasks within projects based on the Kanban method, which will also be used for the development of tasks within this project. Exposure to such methodologies will be a new experience for Natasa providing invaluable insight into R&D project execution. The **Gantt Chart** presents the work plan showing timing of different WPs and project trainings, secondments, events, etc.

3.3.4. Intellectual property rights

The IPR ownership and access rights will be organised, with the default agreement being the IRIS will own the IP generated during this project. Since the IP is mainly generated in WP2 and 3 and then transferred to WP4, it will be used in industrial trials but also sold as a non-exclusive licence to the case study partners with the aim to license it further to sectorial companies post-project. Regarding the Fellow, IRIS will still keep a fellow as inventor on the subscribed IP. Early protection strategies will be in place to enable Open Access Publications for ensuring impact for the Fellow. In the case of the sub-graph matcher patent, co-invented by Dr. Nettleton, the supervisor of the work in this proposal and owned by the Pompeu Fabra University (UPF), Barcelona, its use for research as planned in DATAMINE4.0 is granted for free with a license agreement under fair and reasonable conditions, especially in view of the improvements herein planned that will extent the attractiveness of the patent, will be facilitated to IRIS, should it be needed for post project exploitation.

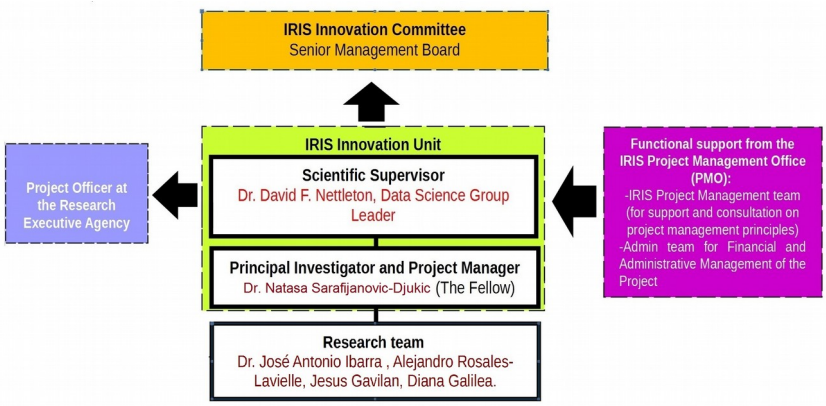

Fig. 3 DATAMINE4.0 Management Structure and the Project Management Procedures

3.4. Appropriateness of the institutional environment (infrastructure)

IRIS will provide the Fellow with a work contract that complies with all Spanish law and with the European Charter of Researchers requirements. In relation to the measures that IRIS will undertake for training, mentoring/tutoring the researcher, IRIS has first of all worked closely with the Fellow in order to agree in the work plan following IRIS methodologies and ISO 9001:9002 and UNE certified procedures for its internal research projects. Moreover, a dedicated WP (WP5) will ensure a systematic and structured approach to training (see 1.2), which will allow for effective operational planning and execution of the scientific, structured and complimentary training activities during the project for the benefit of the Host. In relation to facilities for in-house training, IRIS has its own auditorium and demonstration rooms for knowledge transfer to industry where the Fellow will organize workshops, lectures and conferences related to project. Host will also ensure that Fellow can benefit from their international collaborations, it will be hugely beneficial for both the Fellow and the projects in question, that the Fellow has contact with a number of the very relevant and synergistic European applied and industry/market driven R&D projects that are active in IRIS. DATAMINE4.0 and the training of the Fellow could benefit from synergies with other projects at IRIS to perform also quick exploratory tests regarding the potential of e.g. the in-line process modeling as in the H2020 OPENMIND project or the big data processing (the online streaming repository being built in the H2020 SMART-GRAN project). Further information on the equipment and infrastructure to be used in the project can be found in the Section 5.

Mentoring provided by the **team of engineers and data scientist in IRIS** will be **supplemented by the expertise offered by academic partners**, ensuring a research mentoring vision that is complementary to applied R&D. Based on the its core expertise in chemometric modeling and data science, IRIS has pioneered the Industry 4.0 concept (featured on the End Page), a key integrated environmental development strategy, which is being employed in a number of EU and private B2B development concepts, and which has generated significant interest from industry. DATAMINE4.0 will enable the generation of achievements that will further IRIS's strategic interests (refer to 2.2).

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4. CV of the Experienced Researcher

PROFILE
Natasa Sarafijanovic-Djukic

Country of Birth: Serbia
Date of Birth: 15/09/1974
Current Country of Residence: Switzerland

Languages: English (working proficiency), German (B1 certificate),
French (basic), Serbian (native tongue)

Previous sections showed the excellent match between Natasa's background and the DataMine4.0 project, as well as her international adaptation skills, her learning capacity and strong will. **She was in the top 3% of students at University of Belgrade.** She worked in Ericsson Belgrade as a GSM Engineer for 2 years where she gained an industrial experience in telecommunication networks as well as valuable communication skills with clients. Driven by motivation to further develop and practically apply her mathematical talent and programming skills and by ambition to pursue an international research carrier, she moved to Switzerland to do her PhD studies in Computer/Communication Sciences at EPFL. The main contributions of her PhD thesis are statistical modelling of realistic human mobility and application of these models to the design and development of routing algorithms in mobile networks. This work resulted in 4 scientific articles that she presented in 2 international conferences and in 4 national conferences in Switzerland efficiently communicating her scientific findings to a wider public. During her teaching activities at EPFL she has shown an excellent skill to explain a complex course material to students. After her PhD she has collaborated in several data science projects as a freelancer. During these projects she gained an experience in more recent advancements of machine learning techniques (such as deep learning). Through all her professional and private endeavours she has shown an excellent problem solving abilities and an extraordinary persistence to make these endeavours successful.
DataMine4.0 will allow her a direct involvement within industry projects further developing her scientific skills in applied data modeling and analysis in different domains and further developing her communication and presentation skills, while living a new experience abroad.

EDUCATION AND TRAINING

Main degrees:

09/2003 – 02/2010	**PhD in Computer Science** at the Ecole Polytechnique Fédérale de Lausanne, Switzerland Dissertation: Collaborative Routing in Mobile Partitioned Networks Supervisor: Matthias Grossglauser *Conducted part time with concomitant professional experience described in next section*
10/1994 - 01/2000	**Diploma in Electrical Engineering in Telecommunications** at the Faculty of Electrical Engineering, *University of Belgrade, Serbia*

Additional training:

09/2014 - 02/2015	Data Science Specialization (9 courses + Capstone Project) Johns Hopkins University on Coursera
01/2016 - 03/2016	Machine Learning course (grade:100%) Stanford University on Coursera
11/2015 - 02/2016	Big Data (4 courses) University of California, San Diego on Coursera
12/2015 - 04/2016	Python Specialization (5 courses + Capstone Project) University of Michigan on Coursera
03/2014 - 07/2014	Oracle Certified Professional, Java SE 7 Programmer
10/2014 - 01/2014	Oracle Database SQL Certified Expert
04/2010 - 03/2011	Intensive German Courses (levels, A1, A2 and B1)

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WORK EXPERIENCE

Aug 2012 - present
Data Scientist, Freelancer
Collaboration on different projects in a diversity of commercial and industrial applications, including a project with Gilad Bar Illan (Tsuroni LTD, Israel) from Mar 2015 to Nov 2015 on trading algorithms and social media research and development; development of a system to collect real-time social media data and store it in a MongoDB database; A text classification model for the spam and non-spam messages based on machine learning; An information extraction algorithm of specific information related to trading. Tools used: R programming, MongoDB.

(Note: maternity leave May 2011 – July 2012)

Sep 2003 - Feb 2010
Research (as part of PhD study) and Teaching Assistant, Ecole Polytechnique Fédérale de Lausanne, Switzerland
The results of her research work:

- *Forecast of a future position given the random waypoint mobility model*
- *A stochastic model of human mobility (Heterogenous Random Walk)*
- *COGRAD: an algorithm for learning the mobility model in a distributed way*
- *Island Hopping: a novel efficient routing algorithm for mobile ad-hoc delay-tolerant networks*
- *A mobile-ad hoc network simulator*

As a teaching assistant for a Computer Networking course, she designed, development and supervision of the JAVA distributed multi-threaded applications for the course projects. Tools used: Objective CAML, JAVA, Matlab, shell scripting

(Note: maternity leave January 2004 – September 2004)

Oct 2000 - Aug 2002
GSM Engineer, Ericsson Belgrade, Serbia
Installation, trouble shouting, and developing software fixes for Ericsson GSM mobile networks

PUBLICATIONS (4 peer-reviewed)

2009	Piorkowski, M., **Sarafijanovic-Djukic, N.**, and Grossglauser, M. A Parsimonious Model of Mobile Partitioned Networks with Clustering, COMSNETS 09, Bangalore, India, January 2009. Cited by 194.
2008	Piorkowski, M., **Sarafijanovic-Djukic, N.**, Grossglauser, M., On Clustering Phenomenon in Mobile Partitioned Networks, First ACM SIGMOBILE International Workshop on Mobility Models for Networking Research, Hong Kong, May 2008. Cited by 28.
2006	**Sarafijanovic-Djukic, N.,** Piorkowski, M. and Grossglauser, M., Island Hopping: Efficient Mobility-Assisted Forwarding in Partitioned Networks, IEEE SECON 06, Reston, VA, September 2006. Cited by 131.
2004	**Sarafijanovic-Djukic, N.,** and Grossglauser, M., Last Encounter Routing under Random Waypoint Mobility, NETWORKING 04, Athens, Greece, May 2004. Cited by 34.

Sum of citations (Google Scholar, 12/09/2016): **387**

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Oral	On Clustering Phenomenon in Mobile Partitioned Networks, First ACM SIGMOBILE International Workshop on Mobility Models for Networking Research, Hong Kong, May 2008
	Mobility-Assisted Forwarding in Partitioned Networks, NCCR MICS 5th Scientific Conference, Zurich, Switzerland, October 2006
	Island Hopping: Efficient Mobility-Assisted Forwarding in Partitioned Networks, IEEE International Conference on Sensing, Communication and Networking, Reston, USA, September 2006
Posters	Collaborative Routing in Mobile Partitioned Networks, NCCR MICS 6th Scientific Conference, Zurich, Switzerland, January 2008
	Island Hopping: Mobility-Assisted Forwarding in Partitioned Networks, NCCR MICS Annual Workshop, Neuchâtel, Switzerland, July 2007
	Routing in Delay-Tolerant Networks, NCCR MICS 4th Panel Review and Scientific Conference, Loewenberg, Switzerland, September 2005
	Last Encounter Routing under Random Waypoint Mobility, NCCR MICS 2nd Scientific Conference, Ascona, Switzerland, October 2003

SKILLS AND TECHNIQUES

- Data Science:
- ✓ Probability and statistics, stochastic processes and stochastic modelling
- ✓ Machine learning algorithms: linear regression, logistic regression, support vector machines, decision trees and random forest, naive Bayes, ensemble methods, K-means clustering, dimensionality reduction – PCA.
- ✓ Deep learning: forward neural networks, recurrent neural networks, convolutional neural networks
- ✓ Natural language processing: text classification, information extraction, language modelling, neural machine translation

- Software:
- ✓ Programming: Python, R, Matlab, Java, Objective Caml , C, shell scripting
- ✓ Deep learning: Keras library
- ✓ Relational Databases (Oracle, MicroSoft SQL Server), NoSQL Database (MongoDB)
- ✓ Big data: basics of Hadoop – HDFS, Map/Reduce, Spark, HBase, Hive, Pig
- ✓ Tools: latex, git and github, cvs

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5. CAPACITY OF THE PARTICIPATING ORGANISATIONS

Beneficiary	Investigacio i recerca industrial i sostenible, SL. (IRIS) /Spain
General Description	IRIS is an advanced engineering company that specialises in process monitoring and control, and process optimisation. Our commercial division offers advanced monitoring and control solutions based on: NIR chemical monitoring (handheld device and in-line solutions); hyperspectral imaging solutions for foreign body detection and chemical composition analysis; data management solutions for data processing, decision support, automation and data analysis (big production data). Our research division has three core research lines: Optical and spectroscopy technologies (e.g. MEMs spectrophotometers, fluorescence and Raman) for process monitoring for on-line and PAT applications in food, pharmaceuticals and the process industries Data Processing & Information systems Novel processing technologies (ultrasound, plasma, microwave…) for process intensification and non-thermal decontamination Sustainable processing technologies: waste valorisation strategies, biocompounds and biopackaging for the circular economy The company was established in 2007 and currently employs 48 staff with a turnover of €3.8 M. We are based over two sites (Barcelona and Dublin) that are equipped with engineering workshops (3), food testing and wet chemistry laboratory, optics labs (2), and an electronics/telecommunications lab. IRIS is ISO 9001:2008 certified and also is certified under UNE-EN-ISO 166002: Management Systems R&D&i. IRIS is an industry member if the following sectoral networks: SPIRE, SPIE and SECPHO. IRIS has four main research lines: optical & spectroscopy technologies, data processing & information systems, novel processing, and sustainable processing. In the context of data processing & information systems, the objective is to develop intelligent data management and decision support systems for smart manufacturing, processing and agriculture. The technologies used are: Cloud, Predictive modelling, Data mining and Deep Learning. This research is aligned with the Industry 4.0 framework for the advanced automation of industrial processes. We currently have six H2020 projects in this area (Propat, IbD, Sharebox, Smart-Gran, OpenMind and Preview). IRIS has strong skills and experience in data modeling and statistical models/analysis derived from its involvement in specific data analysis and modeling tasks for Chemometric modeling, as well as analytical and IT monitoring systems in many FP7 and H2020 projects. For example, in the Smart-Gran H2020 project, IRIS is responsible for building a big data processing platform in the cloud for processing data from pharmaceutical production plants distributed all over the world; the Hypera device, designed and built at IRIS, is able to detect foreign-bodies and other non-desired bodies through spatial quality chemometric modeling.
Role and Commitment of key persons (supervisor)	The scientist in charge of the supervision of the project, Dr. David F. Nettleton, holds a bachelor's honours degree in Computer Science (1984) from the University of Wolverhampton (UK), a master's degree in Computer Software and Systems Design (1985) from the University of Newcastle upon Tyne (UK) and a PhD in Artificial Intelligence (2002) from the Technical University of Catalunya (UPC, Spain), the latter with the distinction "excellent cum laude". Since the start of his research career in 1993, Dr. Nettleton has built significant expertise both in academic and applied R&D and is as such perfectly suited to supervise the fellow in the planned research, having been involved in a mentoring role with numerous researchers in the University and in Industry, throughout his career. He has worked in the IT industry for 30 years, starting as an intern in the Atomic Energy Research Establishment in Harwell, UK, followed by 3 years in a major UK software company (Systems Designers) where he participated in high level projects, such as FOLIOS (office automation for the UK Foreign Office). On coming to Spain he worked for Carburos Metalicos (1989-1990) as a technical engineer before embarking on a business startup (DSE, 1990-1993) specialized in Expert System software implementation and consultancy. From 1993-1995 he was contracted by Menarini Laboratories as programmer/analyst to develop the production quality control IT system (complying with ISO 9000-3) and a clinical trials system in Ingres 4GL. From 1995-1997 he embarked again on a business startup (TAD Sistemas), dedicated to data mining consultancy and products (Clementine), carrying out vanguard data modeling/analysis projects in Spain for a major television company (TV3), a retail bank (Bankpyme) and a major health insurance company (Sanitas). After a 1 year stay in Coritel (a major software developer in Spain) as project manager of large teams, David was hired by IBM Global Services as a Business Intelligence Consultant, where he participated in projects for important IBM customers such as Iberia, INTA - National Institute of Aerospace Technology (Spain), General de Depositos (Portugal), Zurich Insurance, among others. At IBM he made extensive use of the powerful IBM Data Mining workbench, Intelligent Miner for Data. After leaving IBM he decided to embark into the academic world (University Pompeu Fabra) as associate lecturer and member of the Web Research Group (led by Dr. Baeza-Yates, VP of Yahoo! Research). At the UPF (2004-2015) he worked on a diversity of projects, aimed at modeling the behavior of users in Internet, especially for search engines. During 2 years (2006-2008) Dr. Nettleton also taught part time (as full lecturer) the final year course of Artificial Inteligence of the Computer Science and Telecomunications degree studies at the Ramon Llull University (URL), also collaborating with the GRSI (Intelligent Systems) research group at the URL. Additionally, he worked for 3 years part time (2010-2013) for the IIIA-CSIC, specializing in data analysis and algorithmic modeling applied to privacy preserving data mining and obfuscation (as member of the ARES research group - Advanced Research on Information Security and Privacy). In recent years he has specialized in graph analysis applied to online social networks (Spanish Ministry Hipergraph and Ares projects) which resulted in a patent application for a graph matcher in colaboration with a researcher from the Katholieke Universiteit Leuven (KU Leuven). Since he joined IRIS, he has successfully collaborated in several European research projects relevant to the present research topic on data mining applied to industrial processes such as aforementioned OpenMind, Sharebox and SmartGran projects.
Key Research Facilities, Infrastructure and Equipment	IRIS is an international company, with its main facility in the Parc Mediterrani de la Tecnologia, a campus of the Polytechnic University of Catalonia in Spain, and a branch facility in the campus of the University College Dublin and has all the necessary **infrastructure and resources** in order to respond to the needs of the project. In particular, IRIS is well equipped with IT infrastructure of hardware and software for data processing and analysis (relational databases, general statistical software packages such as StatGraphics and MatLab and specific statistical software packages for Chemometric modeling (SOLO). Also, for machine learning and AI paradigms IRIS uses Weka as a GUI and as a Java based API library, and Dlib as a C++ based API library for embedded applications. In terms of hardware, almost all the computers in IRIS hace 64 bit processors with 4 or 8Gb memory. A rack configuration of servers and hard disks is also available for parallel processing and big data application. IRIS also has access to academic and research repositories through a partnership with the UPC University Campus

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	where IRIS is located. The team at IRIS is an international team of 48 staff, comprising researchers, engineers, project managers and support staff from Spain, Ireland, France, Italy, Colombia, Cuba, among others but also Serbia which could help in Natasa's insertion. The data science teams in IRIS include Telecomunications and IT Engineers who have a strong track record in FP7 and H2020 of building IT and data processing applications.
Independent research premises?	Yes
Previous Involvement in Research and Training Programmes	Active collaboration with the Universidad Politecnica de Catalunya (UPC), whereby IRIS provides industry mentorship and training to MSc and PhD students. IRIS has successfully mentored 4 MSc. students, as well as 2 PhD students, and a Post-doc student is currently doing her research programme in IRIS through a Marie Curie Fellowship (specializing in Environmental Science). During FP7, IRIS has been involved in close to 40 European projects (SMEs, NMP, KBBE, EraNET, etc.) and is now involved in close to 15 H2020 projects. The scientific supervisor for this project, Dr. Nettleton, has been tutor to two PhD candidates, at the UPF – Pompeu Fabra University, Barcelona, in relation with data analysis research (big data graph processing and recommender systems, respectively), as well as a masters student (big data research) and engineering students and graduated professionals. David also has teaching experience in three Universities with first, second and final year University and engineering students at the Pompeu Fabra University, Ramon Llull University and European University, in terms of magistral (theory) classes, tutorials and practical classes in computer science and artificial intelligence.
Current involvement in Research and Training Programmes	Mentoring/tutoring of research students is something that IRIS is experienced at and committed to. Through our ongoing collaboration with UPC, IRIS currently hosts 2 bachelors student as well as 1 PhD student and a Marie Curie Fellowship in Materials Science. IRIS has a programme for internships and has agreements with a number of local and international schools and institutes for the placement of interns in the company as part of their studies. IRIS also has experienced staff in supervising the creation of research methodologies for setting up and validating experiments. The supervisor is still involved in informal knowledge transfer of experience to students or peers.
Relevant Publications and/or research/innovation products	**In addition to the publications mentioned in Section 2.3.1, further work published is as follows:** **Rosales Lavielle, A**. "Monitorización en línea para el control de calidad a tiempo real", BTA (Barcelona Alimentation Technologies) Congress, 21-24 April, Barcelona. **Nettleton, D.F.**, "Data Mining of Social Networks Represented as Graphs", Computer Science Review 7(I):1-34 · February 2013. **Nettleton, D.F.**, "Information loss evaluation based on fuzzy and crisp clustering of graph statistics". In: Proc. 2012 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE), Brisbane, Australia, June 2012 (pp. 1-8). **Nettleton, D.F**., Salas, J., "A Data Driven Anonymization System for Information Rich Online Social Network Graphs", Expert Systems with Applications Volume 55, 15 August 2016, Pages 87–105. Impact factor 2.24 (SCI 2014/2015) **Nettleton, D.F.**, A Synthetic Data Generator for Online Social Network Graphs, Soc. Netw. Anal. Min. (2016) 6: 44. doi:10.1007/s13278-016-0352-y **Nettleton, D.F**., Salas, J., Approximate Matching of Neighborhood Subgraphs — An Ordered String Graph Levenshtein Method, Int. Journal of Uncertainty Fuzziness and Knowledge-Based Systems 24(03):411-431 · June 2016. Impact factor 0.619 (SCI 2013) **Hypera** (IRIS project)ID H2020-SMEINST-2-2016-2017, "Affordable Hyperspectral Imaging System for on-line foreign body detection and chemical composition analysis", https://ec.europa.eu/easme/en/sme/8301/affordable-hyperspectral-imaging-system-line-foreign-body-detection-and-chemical **Sharebox** (IRIS project) H2020 SPIRE-06-2015 Secure Management Platform for Shared Process Resources http://sharebox-project.eu/ **Smart-Gran** (IRIS project) H2020-SMEINST-2-2014, Development and Commercialisation of a Self-Guided Fluidised Bed Granulation Solution, http://cordis.europa.eu/project/rcn/196401_en.html

Partner Organisation	**University of Skövde, Sweden (secondment 1)**
General description	**The University of Skövde** is a modern university located in the south of Sweden and has approximately 11000 students and 414 educational staff, structured in 5 schools. The university has many collaborations with international and local companies, including mentoring programs and visiting scholars, and is located close to the Gothia Science Park. The School of Informatics has a close relationship to society, well established infrastructure and a stable financial base, giving the conditions to engage in both Swedish and international education as well as research. Research at the School of Informatics covers topics such as artificial intelligence, cognitive/interactive systems, distributed real-time systems, information systems, software systems and media, technology and culture. Their research received high ratings in the ARC13 research evaluation study. The school´s research is frequently funded by European Union, The Knowledge Foundation, Vinnova and industry partners. The Skövde Artificial Intelligence Lab (SAIL) conducts research within applied AI in close collaboration with industry. Current research activities are focused on reasoning under uncertainty, deep learning, visual analytics, transparent decision support, data privacy and recommender systems. In terms of education, SAIL is responsible for the University's master program in Data Science, the first of its kind in Sweden. The SAIL research group currently has 2 full professors, 3 associate professors, 5 senior lecturers, 1 post-doc and 7 PhD students. Their main role in the project will be to second IRIS in the training of the fellow regarding data science research and big data. They will bring an academic mentoring capacity in the project and complement IRIS's more applied research focus.
Key Persons and Expertise (supervisor)	**Prof. Vicenc Torra**. Leader of Skövde Artificial Intelligence Lab. Computer Science at Technical University of Catalunya (1991) and PhD Computer Science (1994) at Technical University of Catalunya. Previous positions. Assoc. Prof. Research Track (1999-2014) at the Artificial Intelligence Research Institute (IIIA-CSIC), Assoc. Prof. (1991-1999) at the Universitat Rovira i Virgili. Visiting researcher at the University of Tsukuba (Japan). Since 2014 he is Full Professor of Information Technology. His research interests include approximate reasoning (fuzzy sets, belief functions and imprecise probabilities), data privacy (statistical disclosure control and privacy preserving data mining), aggregation operators and data fusion, decision making and game theory, and have led to the publication of over 100 scientific papers. Prof. Torra is currently organizing the *4th Swedish Workshop on Data Science (SweDS 2016, Skövde, Sweden*, the *9th Intl. Summer School on Aggregation Operators (AGOP 2017, Skövde, Sweden)* and the *14th Intl. Conf on Modeling Decisions for Artificial Intelligence (MDAI 2017, Kitakyushu, Japan)*.

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	Selected projects in which Prof. Torra has been coordinator and/or participated as senior researcher: SMASH: Multi-agent systems and their application in hospital services (1996-2001) Funded by: CICYT (Spanish government) (Contract: TIC96-1038-C04-04); STREAMOBILE: Streaming of multimedia content on mobile devices with micropayments (2002-2004) Funded by: CICYT (Spanish government) (Contract: TIC2001-0633-C03-02); e-AEGIS: Electornic shield for balancing consumer privacy and transaction security in the information society: privacy in data bases (e-AEGIS/DB-privacy) (2007-2012) Funded by: MEC (Spanish government) (Contract: TSI2007-65406-C03-02); ARES: Advanced Research Team into Information Security and Privacy (2007-2012); Funded by: MEC (Spanish government) (Contract: 21302); co-Privacy: Sustainable privacy for a sustainable information society (co-privacy/DBprivacy) (2012-2015) Funded by: MEC (Spanish government) (Contract: TIN2011-27076-C03). He will act as supervisor of the fellow.
Key Research facilities, infrastructure, equipment	The research group is currently active in multiple data science and data mining projects, including big data analytics, data fusion, distributed decision making, reasoning about uncertainty, deep learning (including Boltzmann machines and belief networks), visual analytics, recommender systems and data privacy. Many of these projects are being carried out in collaboration with leading edge industrial companies. No specific infrastructure is required for data science and data mining projects except statistical software packages, machine learning software libraries and case study data which are available.
Previous and Current Involvement in Research and Training Programmes	Current projects of the SAIL Research Group include: "A smarter society through Big Data analytics" financed by The Knowledge Foundation; "Big Data Fusion (BISON)" financed by The Knowledge Foundation with collaborating partners Advectas AB, AstraZeneca AB, Huawei Technologies Sweden AB and Takara Bio Europe AB; "HMI for Distributed Decision-Making" financed by the Swedish Governmental Agency for Innovation Systems (VINNOVA) with collaborating partners Saab Aeronautics AB; "New Opportunities through Visual Analytics for Big Data" financed by The Knowledge Foundation with collaborating partners Combitech AB and Takara Bio Europe AB. All of these projects are relevant to the present research topic on data science / data mining case studies and which cover aspects of big data analytics, data fusion and knowledge representation and processing. In terms of teaching the key person involved (Prof. Torra) is a full professor with teaching responsibilities in computer science and data science. Prof. Torra has been thesis director of 6 Ph.D. students and the Research Group currently has 1 post-doc and 7 PhD students and have recently initiated a Masters degree course in Information Science.
Relevant Publications and/or research/innovation product	**Torra, V.** and Narukawa, Y. (Eds.) (2015). Modeling Decisions for Artificial Intelligence: 12th International Conference, MDAI 2015, Skövde, Sweden, September 21-23, 2015, Proceedings. Paper presented at *12th International Conference on Modeling Decisions for Artificial Intelligence, MDAI 2015*, Skövde, Sweden, September 21–23, 2015. **Torra, V**. (2015). Transparency and Disclosure Risk in Data Privacy. In Proceedings of the Workshops of the EDBT/ICDT 2015 Joint Conference (EDBT/ICDT), Brussels, Belgium, March 27th, 2015, pp 246. Abril, D., **Torra, V.**, and Navarro-Arribas, G. (2015). Supervised Learning Using a Symmetric Bilinear Form for Record Linkage. Information Fusion 26, 144–153. **Nettleton, D.F.**, Sáez-Trumper, D., **Torra, V.**, "A Comparison of Two Different Types of Online Social Network from a Data Privacy Perspective", Lecture Notes in Artificial Intelligence, Vol. 6820, pp 223-234, 2011. **Nettleton, D.F.**, **Torra, V.**, "Data privacy for simply anonymized network logs represented as graphs - considerations for graph alteration operations", International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems, December 2011, Vol. 19, No. supp01: pp. 107-125.

Partner Organisation	**KU Leuven, Belgium (secondment 2)**
General description	The KU Leuven is a renowned research-oriented university, carrying out fundamental and applied research in all academic disciplines with a clear international orientation. It was founded in 1425 and has more than 30000 students and over 4400 full-time researchers. KU Leuven's Department of Computer Science consists of seven research groups has 30 professors, 300 master students and 150 doctoral students. One of its research groups, the lab for Declarative Languages and Artificial Intelligence (DTAI) is participating in this proposal. Its subgroup on machine learning has four faculty members (Luc De Raedt, Hendrik Blockeel, Bettina Berendt and Jesse Davis), around 5 post-docs and over 25 doctoral students. The ML group is internationally renowned for its research on machine learning and data mining for structured data (e.g., graphs, relational databases, etc.), statistical relational learning, probabilistic programming, and constraint programming among others. Next to fundamental research, DTAI applies its developed techniques to problems in health-care, prognostics, robotics, computer vision and mechatronics, natural language processing, sports and the life sciences. The ML group's main role in the project will be to support IRIS in training the fellow on applying data mining to graph data. They will bring an academic mentoring capacity to the project and complement IRIS's more applied research focus. Dr. Anton Dries of the ML Research Group has previously collaborated with Dr. David Nettleton, and they are co-authors of the previously mentioned graph matching patent application, which we hope to exploit in this proposal.
Key Persons and Expertise (supervisor)	**Prof. dr. Jesse Davis** is an associate professor of Computer Science at KU Leuven. He received his PhD from the University of Wisconsin – Madison (supervisor: David Page) in 2007 and spent three years as post-doctoral research at the University of Washington (supervisor: Pedro Domingos). He joined the faculty at KU Leuven in 2010. Jesse's research interests include machine learning and data mining for relational data, statistical relational learning, transfer learning, probabilistic graphical models, and empirical evaluation. Furthermore, he is interested in challenging applications in health care, sports, and natural language processing. According to Google Scholar, he has been cited over 2500 times. He has been (co) PI on numerous grants (national, CHIST-ERA) and received an EU FP7 Marie Curie Career Integration Award. He is an action editor for the Machine Learning Journal and was program co-chair for ILP 2014. He will act as co-supervisor of the fellow. **Dr. Anton Dries** is post-doctoral researcher in the ML group. He obtained a PhD in Computer Science at KU Leuven in 2010, for his thesis: "Data streams and information networks: a knowledge discovery perspective". He has more than 10 years of experience in Data, Graph and Stream Mining. He has co-authored more than 30 peer-reviewed articles and has 1 patent application on graph matching (as a co-author with Dr. David Nettleton). His research mainly focuses on data mining, machine learning and probabilistic/constraint programming. He will act as co-supervisor of the fellow.
Key Research facilities, infrastructure,	The research group has wide experience and knowledge in data mining and machine learning. The group currently focuses on applications in the life sciences (especially chemo- and bio-informatics), in constraint-based data mining and inductive databases, and statistical relational learning (combining probabilistic models with logic). The group is actively developing several

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equipment	important software systems such as the probabilistic programming language ProbLog (see https://dtai.cs.kuleuven.be/software/ for a full list of open-source software).and several systems for learning with probabilities and/or structured data (e.g. ProbFOIL, LearnSDD, ACE, ...). Furthermore, the group has access to the Flemish Super Computer for running large scale experiments.
Previous and Current Involvement in Research and Training Programmes	Members of the group have been involved as coordinators or principle investigators in many European projects, including the ESPRIT III and IV projects ILP (Inductive Logic Programming I and II), EU FP7 IST FET project BISON, and the EU FP7 FET project ICON, among others. Current projects include: FWO-TBM Mobile monitoring of joint loading profiles in persons with degenerative hip and knee problems MMJLP; ERC Advanced Grant Synthesizing Inductive Data Models (SYNTH); FWO Updates and Provenance in Data and Knowledge; SMGIS; KUL-C3 Knowledge-Based Software Engineering: IDP4; FWO MERCS: Efficient modeling of big data with multidirectional ensembles of decision trees, CHIST-ERA ReGROUND: Relational ground through affordance learning; IWT-SBO Hypermodelling strategies on multi-stream time-series data for operational optimization HYMOP. The team is involved in teaching classes on machine learning, data mining, and big data. Furthermore, the group offers a class on Data Science and another on Coping with Big Data to members of industry and the wider university community. The group supervises a large number of master theses every year, in addition to mentoring PhD students and post-doctoral researchers.
Relevant Publications and/or research/innovation product	De Raedt, L.**, Dries, A.,** Thon, I., Van den Broeck, G., Verbeke, M. "Inducing Probabilistic Relational Rules from Probabilistic Examples", International Joint Conference on Artificial Intelligence (IJCAI'15) Guns, T., **Dries, A.,** Tack, G., Nijssen, S., De Raedt, L., "MiningZinc: A modeling language for constraint-based mining", International Joint Conference on Artificial Intelligence (IJCAI-13) **Nettleton, D.F., Dries, A.**, "Local Neighbourhood Sub-graph Matching Method", International and European Patent Aplication PCT/ES2014/065505, 2013/2014. **Nettleton, D.F.**, Torra, V., **Dries, A.**, "A comparison of clustering and modification based graph anonymization methods with constraints", International Journal of Computer Applications 95 (20), 2014 Van Haaren, J., **Davis, J.**, Ben Shitrit, H., Fua, P. "Analyzing Volleyball Match Data from the 2014 World Championships Using Machine Learning Techniques" ACM Conference on Knowledge Discovery and Data Mining (KDD 2016) Bekker, J., **Davis, J.**, Choi, A., Darwich, A., Van den Broeck, G. "Tractable Learning for Complex Probability Queries" Neural Information Processing Systems (NIPS 2015)

6. ETHICAL ISSUES

During the preparation of this proposal, in keeping with the required Ethics Appraisal Procedure, an Ethics Self-Assessment was performed with the completion of the Ethical Issues Table (EIT) in the Part A. It can be concluded that this project does not involve any **sensitive ethics issues**.

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7. LETTERS OF COMMITMENT

KU LEUVEN

DEPARTMENT OF COMPUTER SCIENCE
CELESTIJNENLAAN 200A BOX 2402
3001 HEVERLEE, BELGIUM

To Whom It May Concern:

We hereby submit a letter of intent to host a secondment during the Marie Curie Fellowship of Dr. Natasa Serafijanovic-Djukic should this later be approved for funding.

We envisage that the principal terms of the proposed training would be substantially as follows.

1. Hosting of Dr. Natasa Serafijanovic-Djukic, for about 2 months (using his/her Marie Curie travel grant planned for such secondment) in the Machine Learning Research Group of the University of Leuven (KU Leuven).
2. Theoretical and hands on training on applied Data Science, tutored by Prof. Jesse Davis and Dr. Anton Dries on case studies from the DataMine4.0 project as well as other related projects and techniques, such as matching of data represented as graphs, in collaboration between IRIS and University of Leuven (KU Leuven).
3. Co-authorship of any publication that may arise from the joint work

Dr. Natasa Serafijanovic-Djukic's profile fits perfectly our current projects and is outstanding, we therefore hope to be given the opportunity for this collaboration.

Sincerely,

Jesse Davis
Associate Professor
KU Leuven

TEL. + 32 16 32 77 00 FAX + 32 16 32 79 96
www.cs.kuleuven.be

MEMBER OF ASSOCIATIE KU LEUVEN

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UNIVERSITY
OF SKÖVDE

2016-09-12

To whom it may concern,

We hereby submit a letter of intent to host a secondment during the Marie Curie Fellowship of Dr. Natasa Sarafijanovic-Djukic should this later be approved for funding.

We envisage that the principal terms of the proposed training would be substantially as follows.

1. Hosting of Dr. Natasa Sarafijanovic-Djukic, for about 2 months (using his/her Marie Curie travel grant planned for such secondment) in the Skövde Artificial Intelligence Lab of the University of Skövde.
2. Theoretical and hands on training on applied Data Science, tutored by Dr. Vicenc Torra on case studies from the DataMine4.0 project as well as other related projects in collaboration between IRIS and University of Skövde
3. Co-authorship of any publication that may arise from the joint work

Dr. Natasa Sarafijanovic-Djukic's profile fits perfectly our current projects, we are very much interested in the possibility of collaborating with IRIS, and we therefore hope to be given the opportunity for this collaboration.

Sincerely,

Prof. Vicenc Torra

signature.

UNIVERSITY OF SKÖVDE

MAILING ADDRESS	VISITS	PHONE	FAX	E-MAIL	INTERNET
P.O. Box 408, SE-541 28, Skövde, SWEDEN	Högskolevägen	+46 500 44 80 00	+46 500 41 63 25	info@his.se	www.his.se

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ENDPAGE

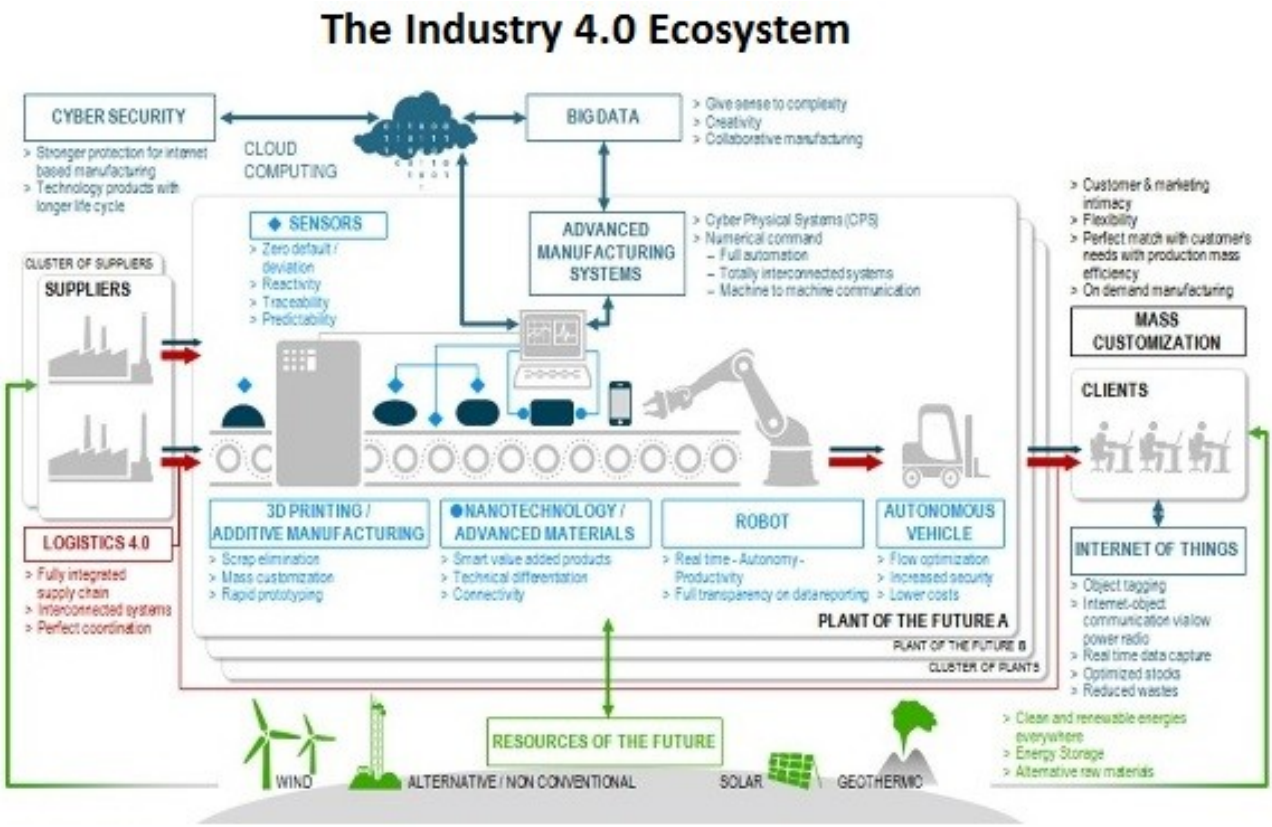
MARIE SKŁODOWSKA-CURIE ACTIONS

Individual Fellowships (IF)
Call: H2020-MSCA-IF-2016

PART B

„DATAMINE4.0"

This proposal is to be evaluated as:
Standard EF



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European Commission

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